# ASTR 302 Fall 2026
# Characteristics of Spectrographs



IGRINS Spectrograph (Park+ 2014)
SPIE Paper

# Agenda – Oct 5, 2026

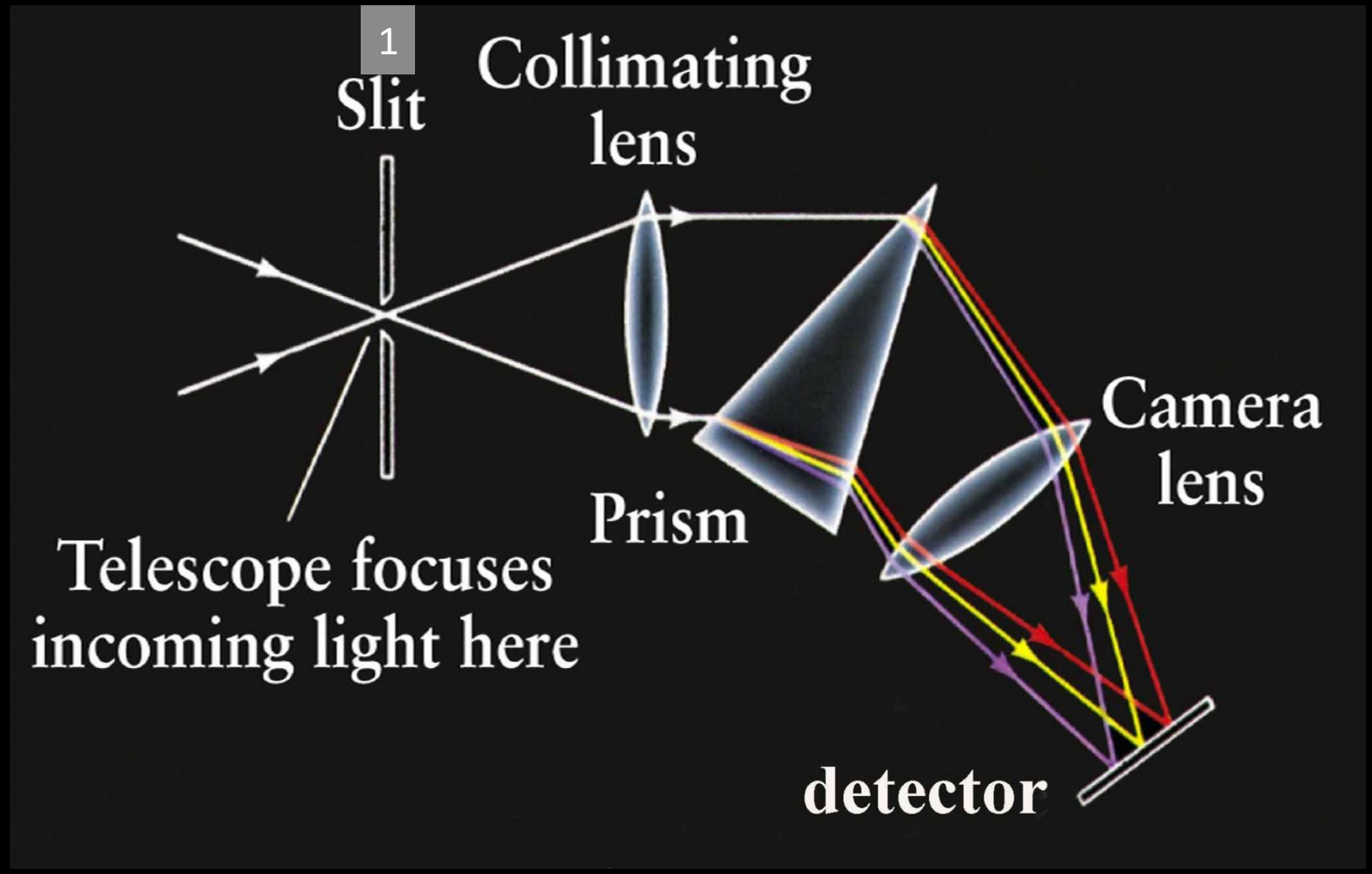
- Weekend Recap
- Photometry Lab due Friday October 9th
- Spectrographs (Dispersers)

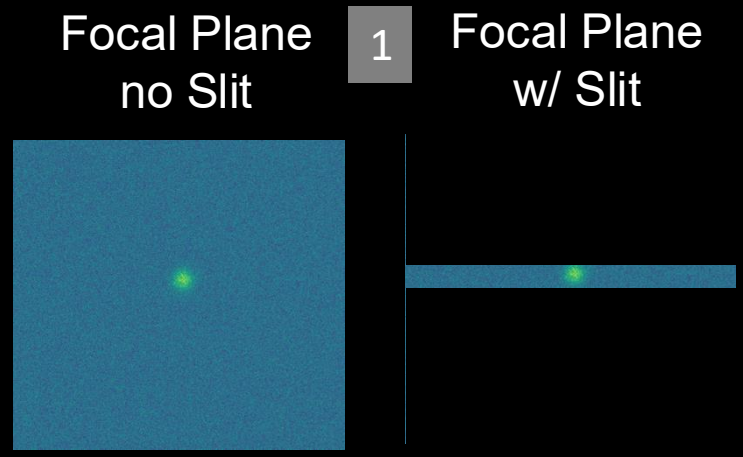
**Recommended Reading: Birney Chapter 12**

# Weekend RECAP!

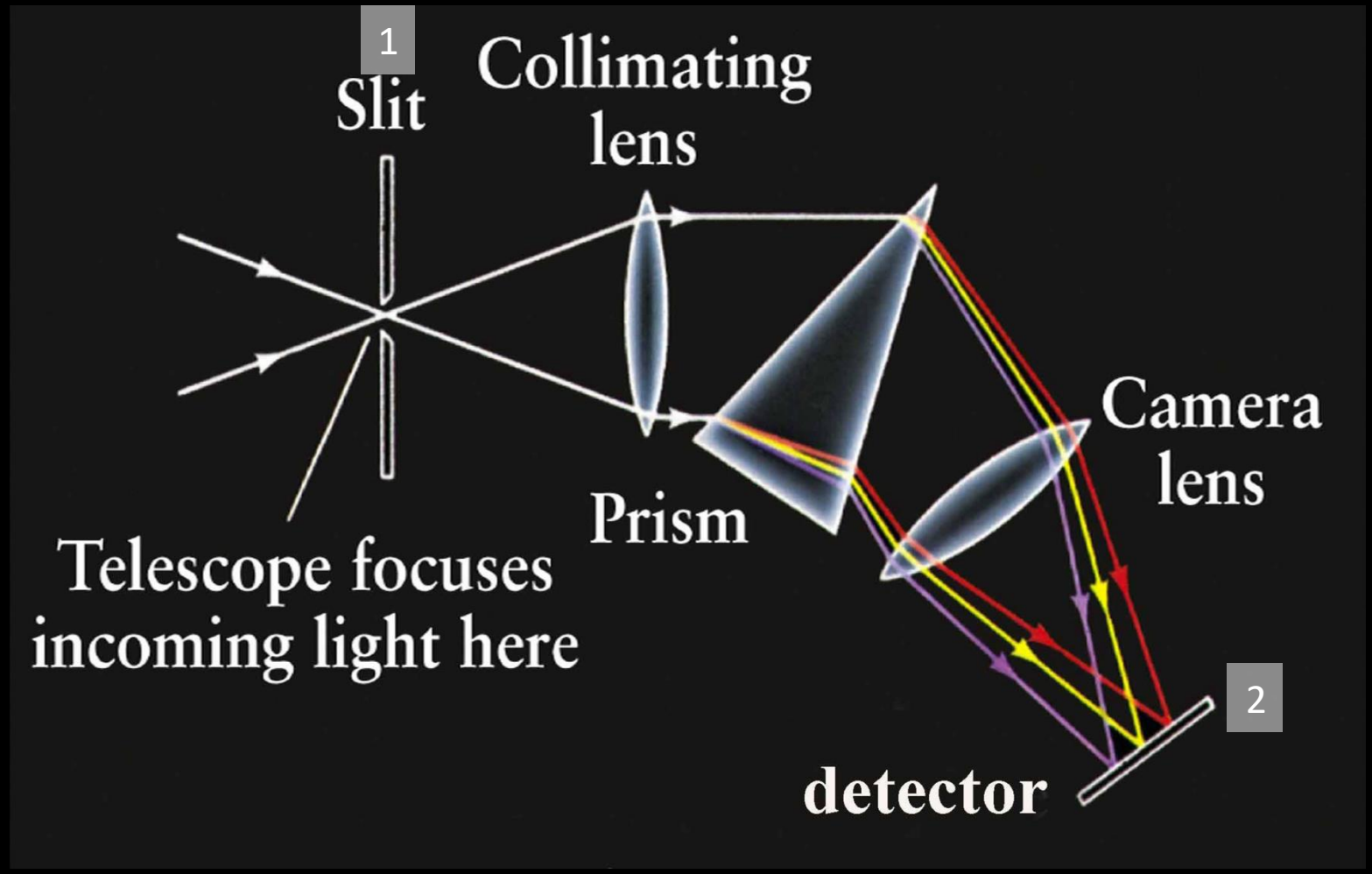


# Components of a Spectrograph

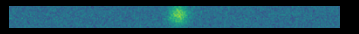


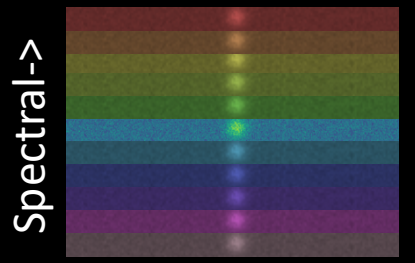
# Components of a Spectrograph




Focal Plane w/ Slit 1



Detector 2



# Characteristics of Spectrographs

**Dispersion Power –** How much different wavelengths are spread out. This is determined by whatever dispersive element(s) is used.

**Resolution –** How well your instrument can measure wavelength differences. This is determined by all your optics and detector together. This is usually what an observatory or instrument manual would report.

# Types of Dispersive Elements

- **Prism** (Transmissive)
- **Diffraction Grating** (Transmissive or Reflective)
- **Grism** (Prism + Grating)

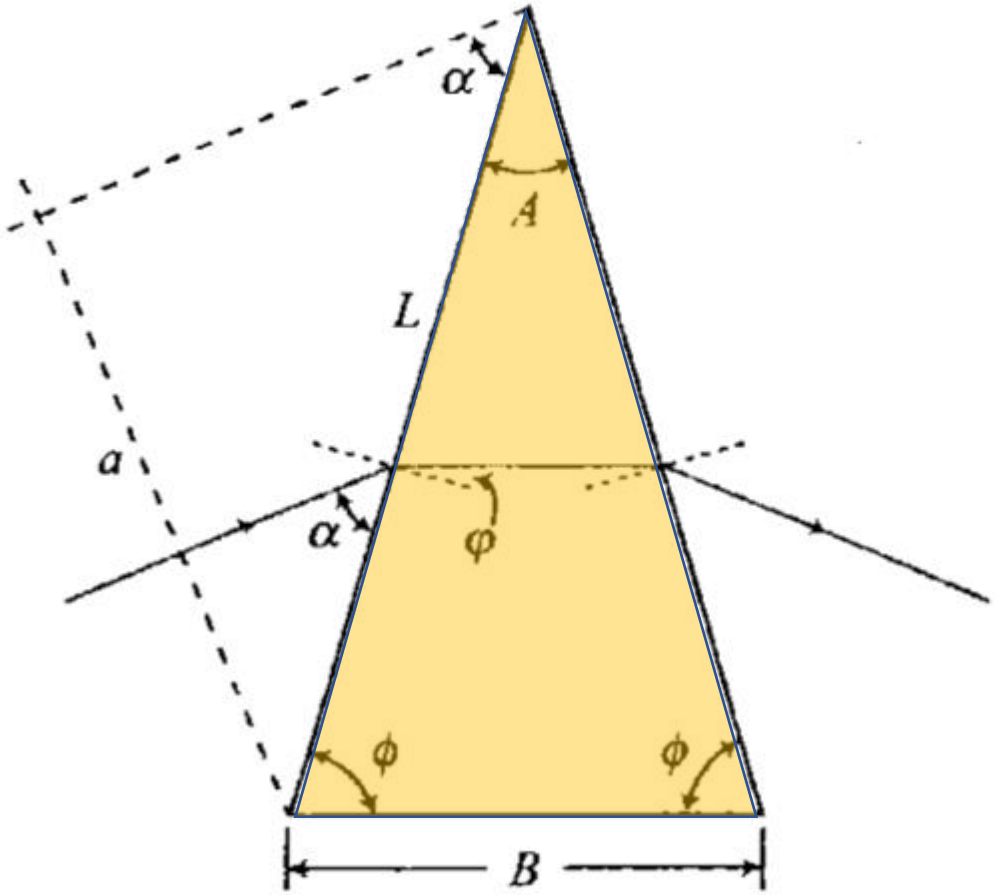
# Types of Dispersive Elements: Prisms

- **Prisms** disperse light by capitalizing on the fact that refraction is a wavelength dependent phenomena.

- The dispersion power of a prism depends on the shape of the prism and the material's index of refraction n($\lambda$):

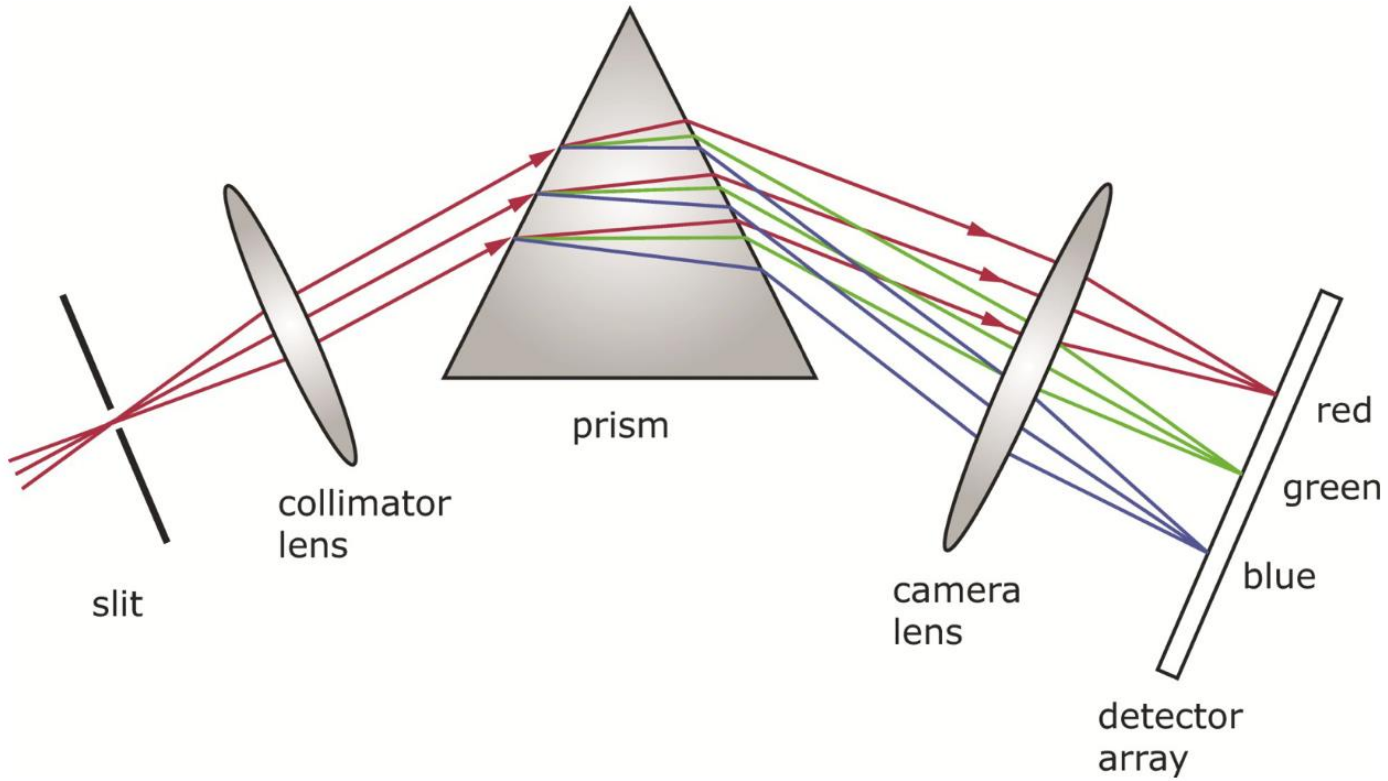
$$\frac{d\theta}{d\lambda} = \frac{B}{a}\frac{dn}{d\lambda}$$

- Thicker prisms have higher dispersion power.




Derivation in Chapter 12 in Obs Book

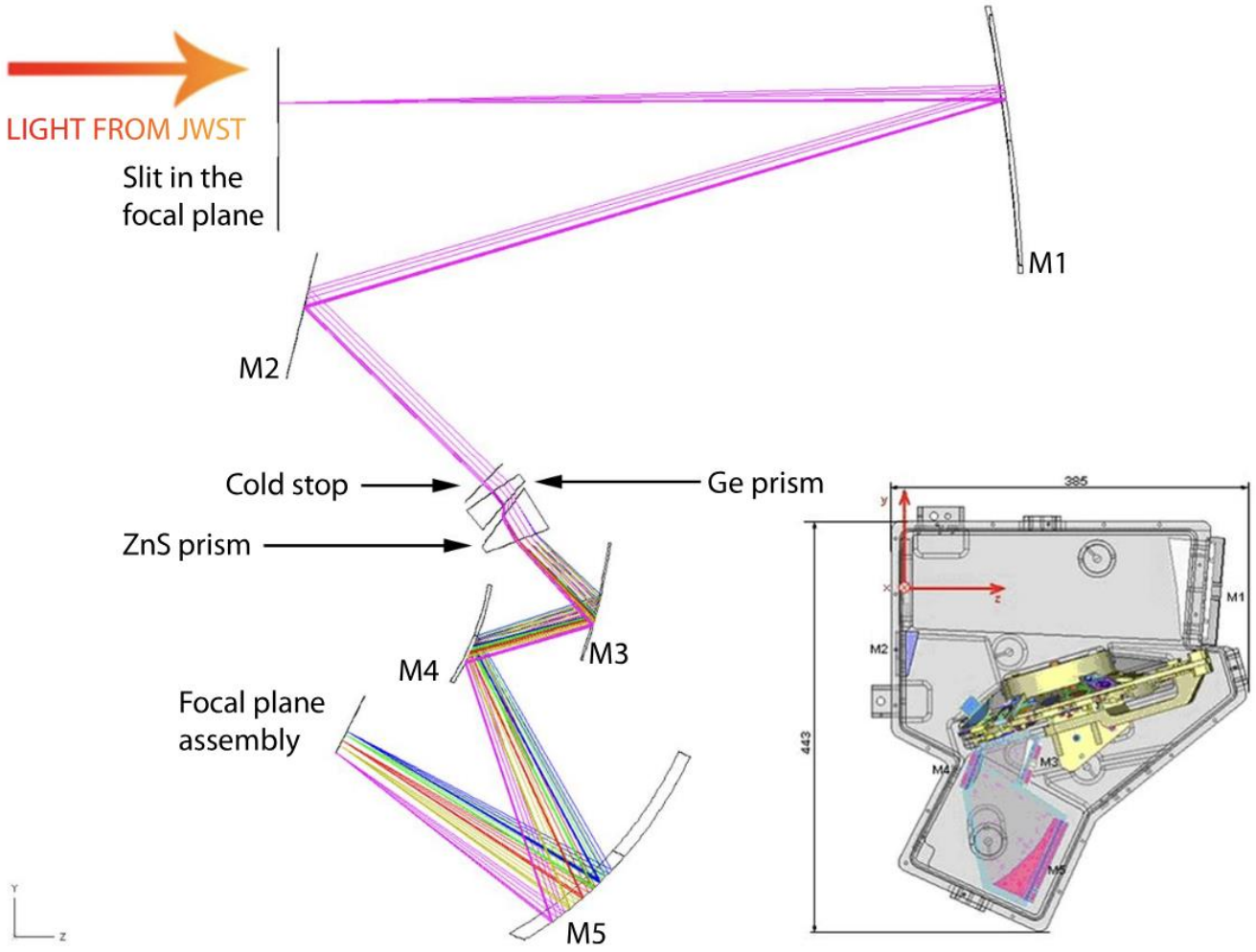
# Types of Dispersive Elements: Prisms




If index of refraction n($\lambda$) changes with wavelength, so does the dispersion power.

$$\frac{d\theta}{d\lambda} = \frac{B}{a}\frac{dn}{d\lambda}$$

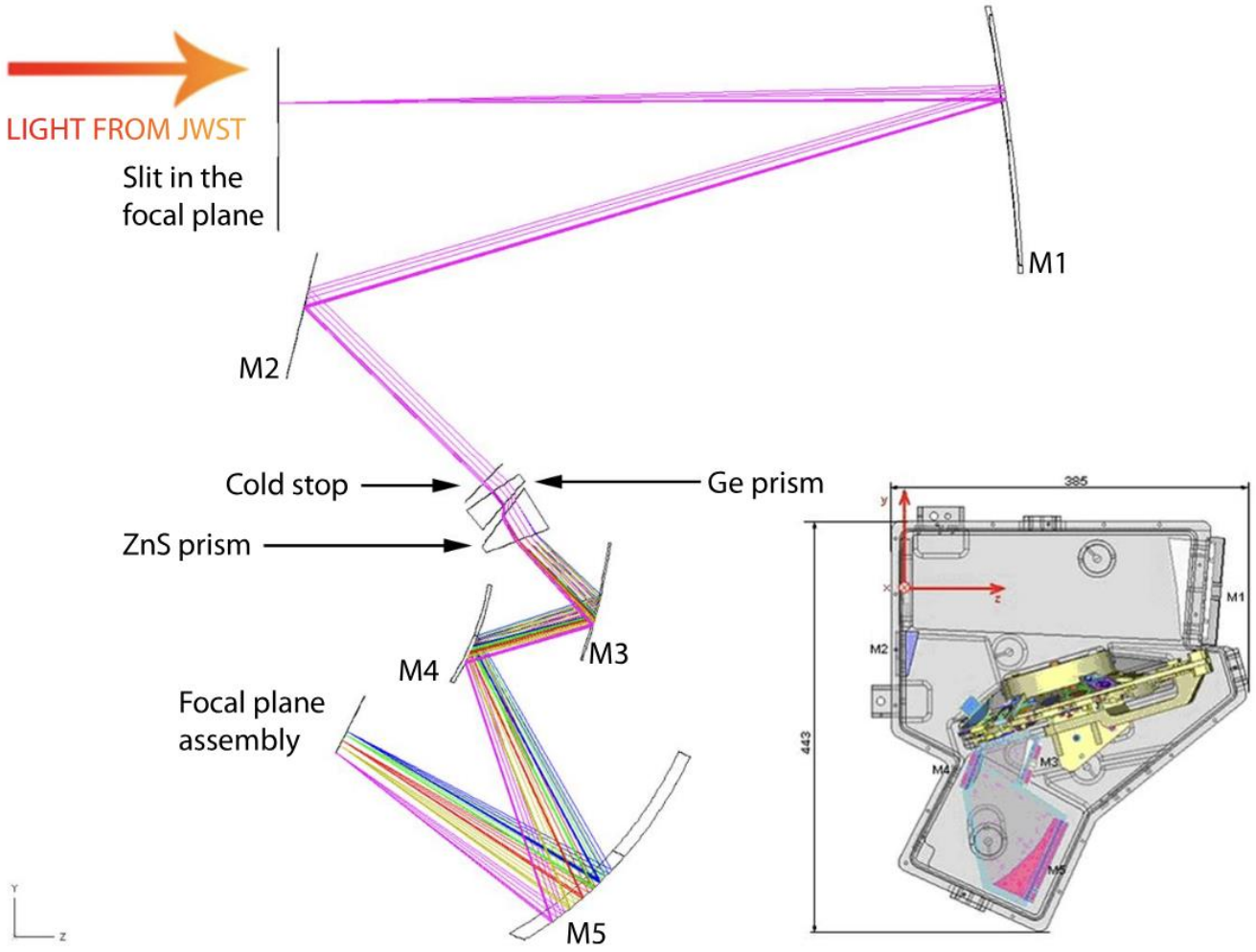
# Prism Example: JWST/MIRI




- Prisms can be stacked or placed in series to increase resolution power.
- Several prisms can be used to keep the telescope beam relatively straight.
- This makes it easy to swap between imaging and spectroscopy using the same detector

Fischer+ 2008

# Prism Example: JWST/MIRI

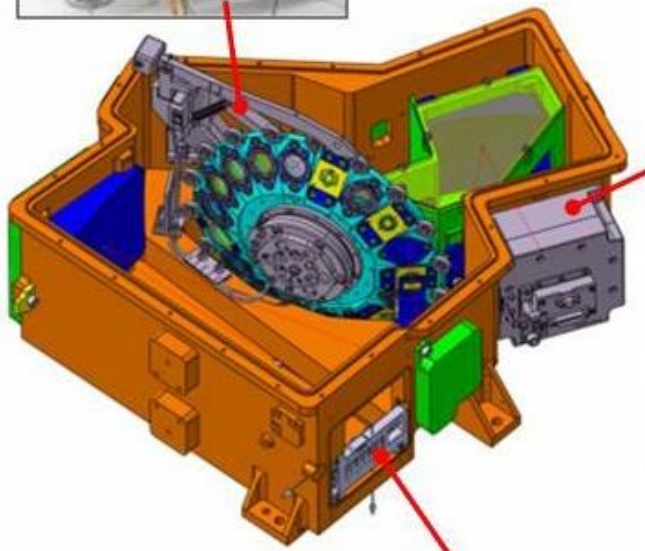







**18 Station Filter Wheel**

- Filters (10), LRS prism (1), chronographic diaphragms and filters (4), open (1), closed+PAR(1), lens (1)





**Focal Plane Module**

- 6.7K operating temperature
- 1024 x 1024 Si:As detector array thermally and electrically isolated from housing

**Coronagraphic masks**

- 4QPM (3)
- Lyot mask (1)



Fischer+ 2008

Jdocs Website (Arizona team provided)

# Prism Example: JWST/MIRI

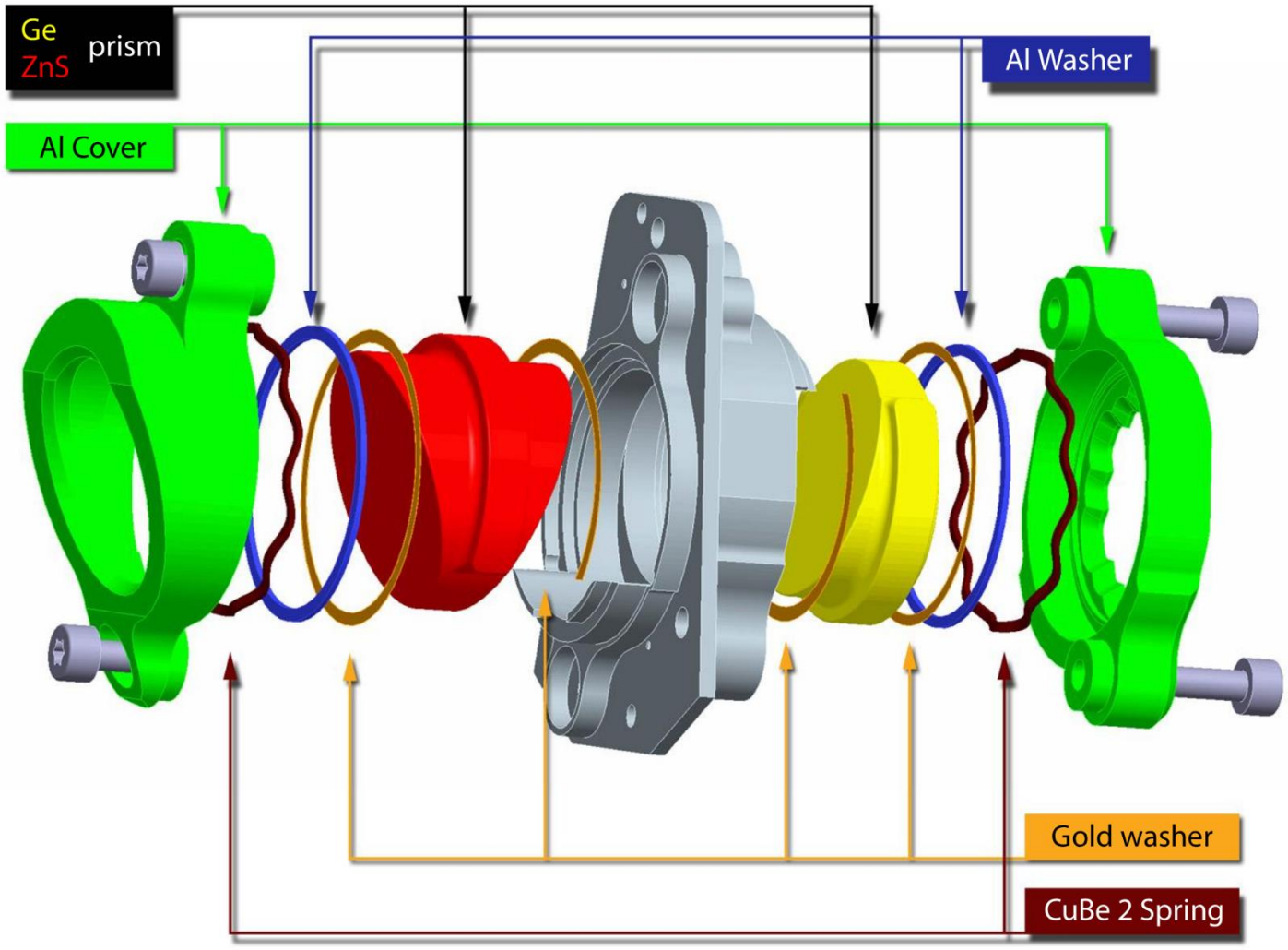



Figure 3. An exploded view of the LRSDPA and its components.

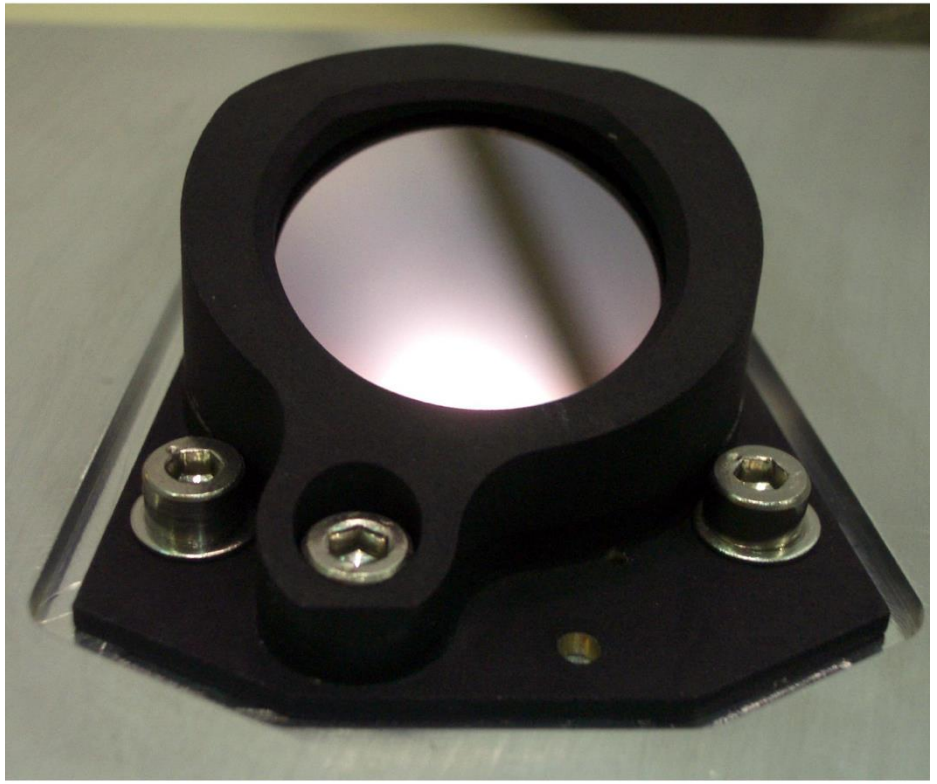


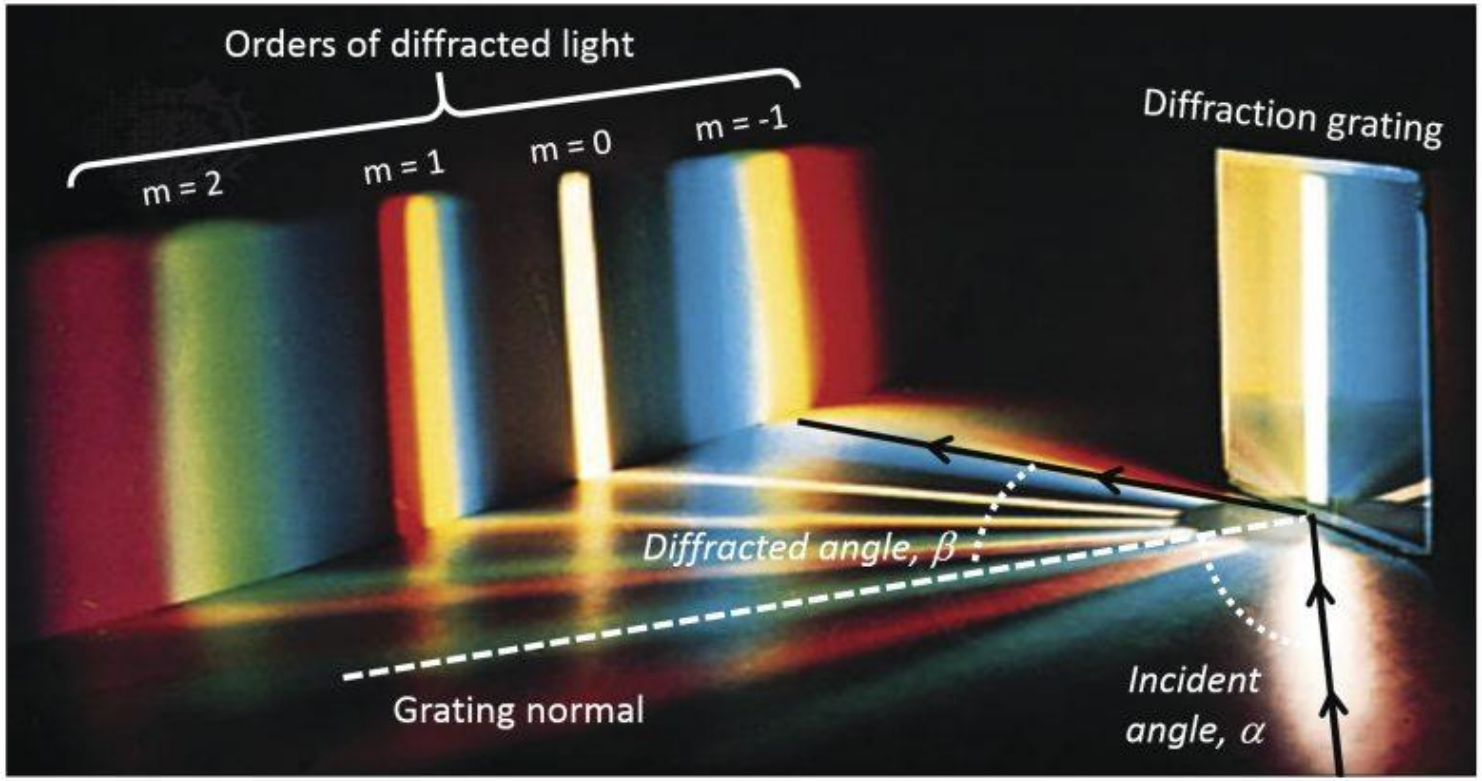
Figure 7. The flight model of the LRSDPA (view onto the ZnS side).

Fischer+ 2008

# Types of Dispersive Elements: Gratings

- **Diffraction Gratings** disperse light into separate wavelengths using diffraction. Multiple grooves are used to increase dispersion power.

- Gratings can be reflective or transmissive.

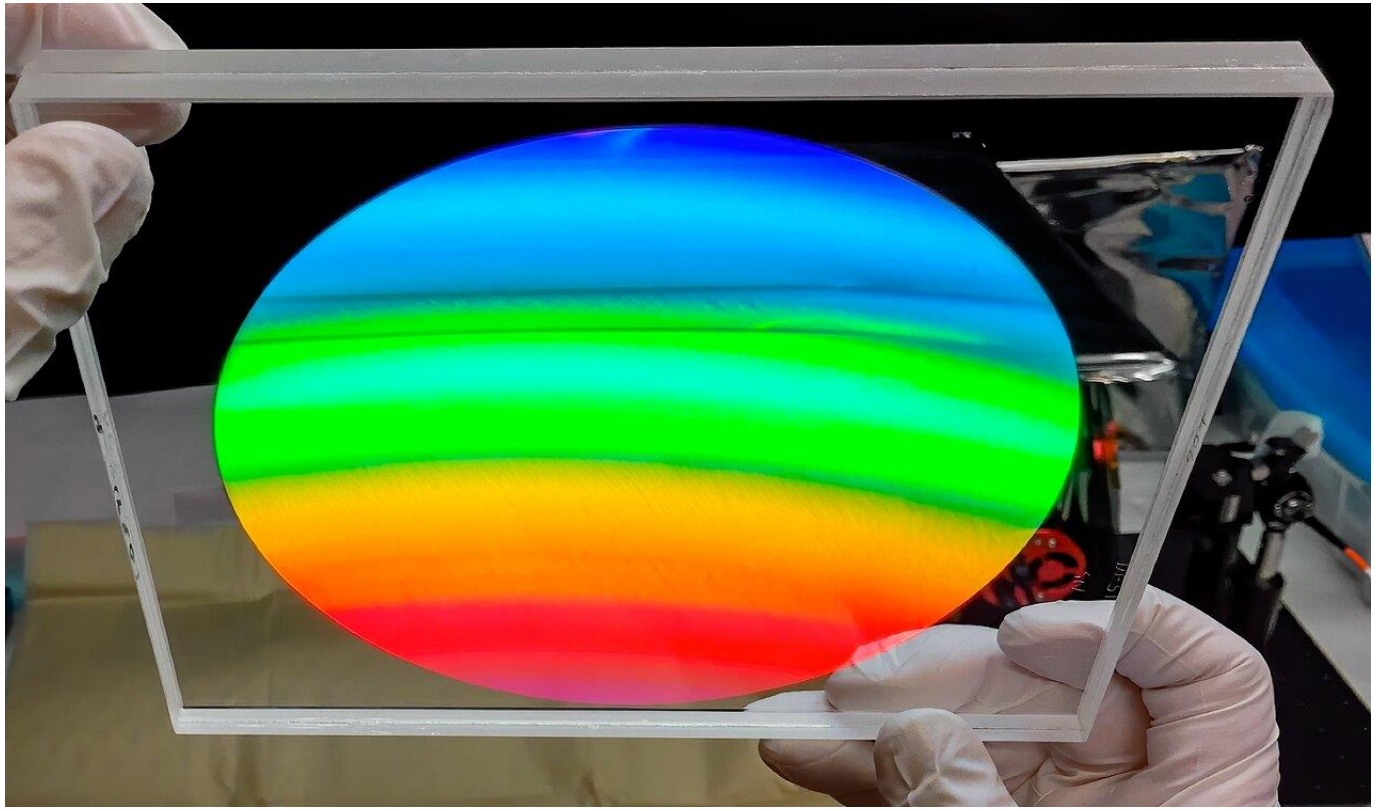



Reflective Grating: Newport

# Types of Dispersive Elements: Gratings

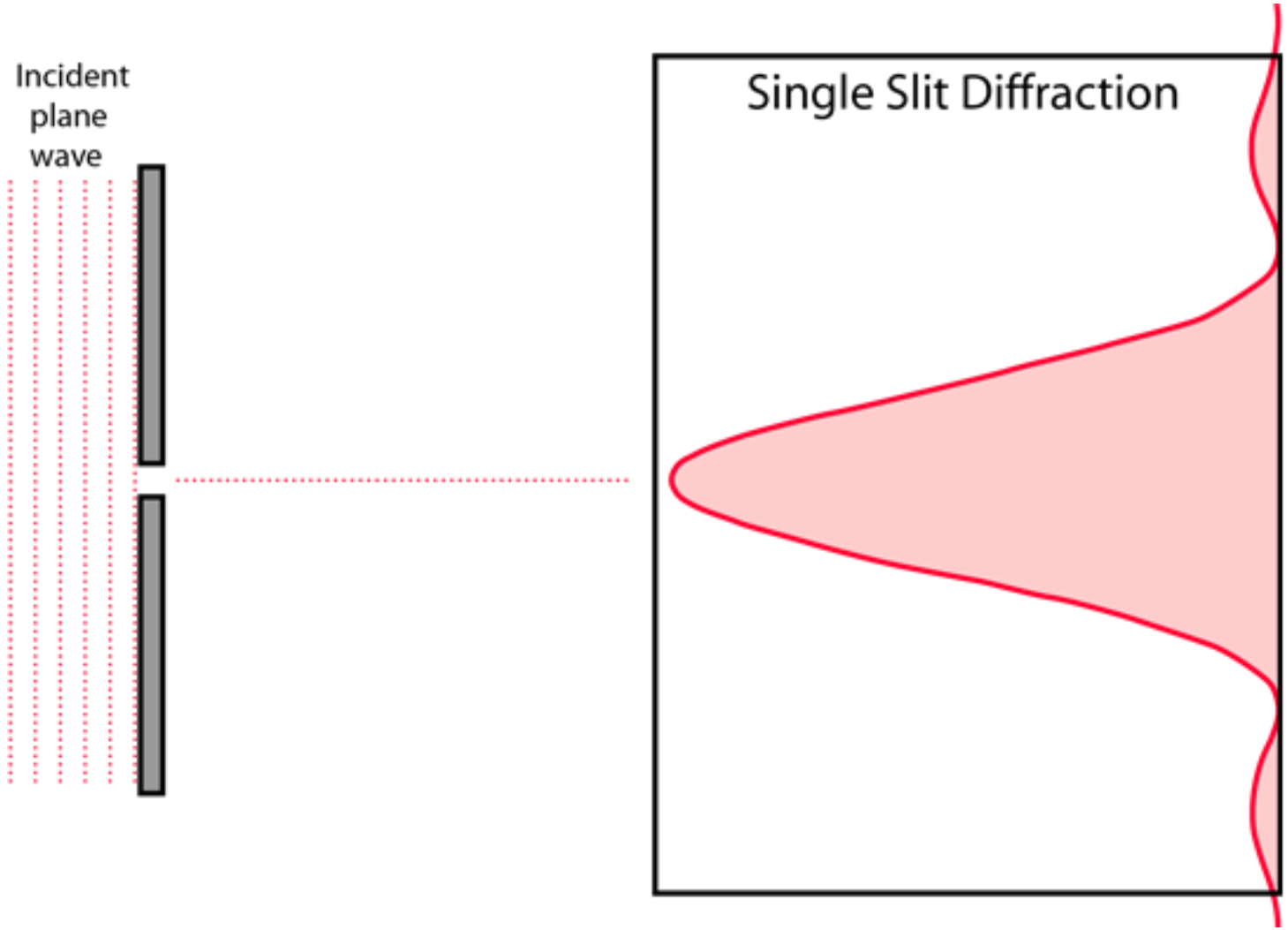
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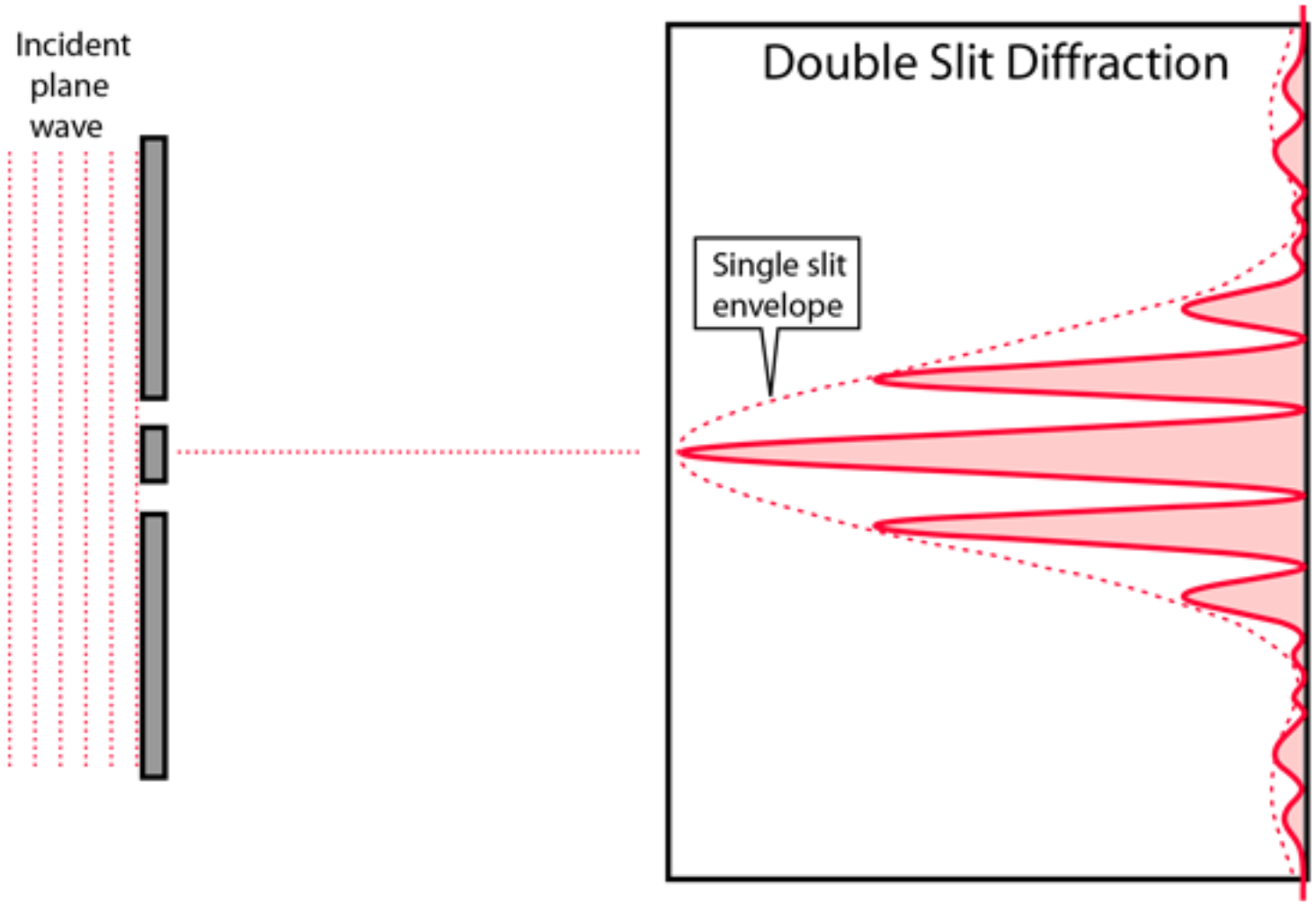
Transmissive Grating: VPH, DESI (NOIRLab)

# Monochromatic Diffraction



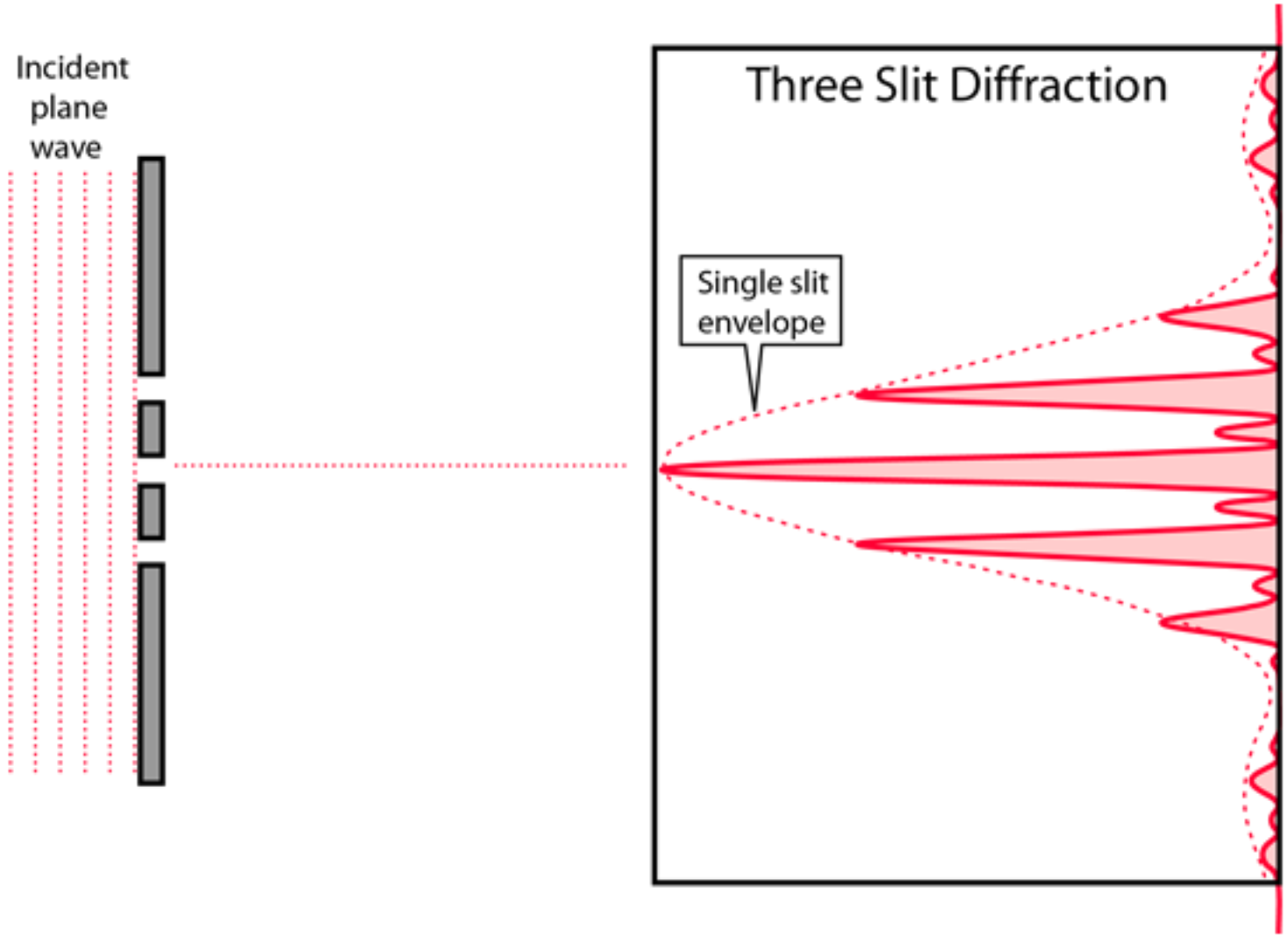

Hyperphysics

# Monochromatic Diffraction



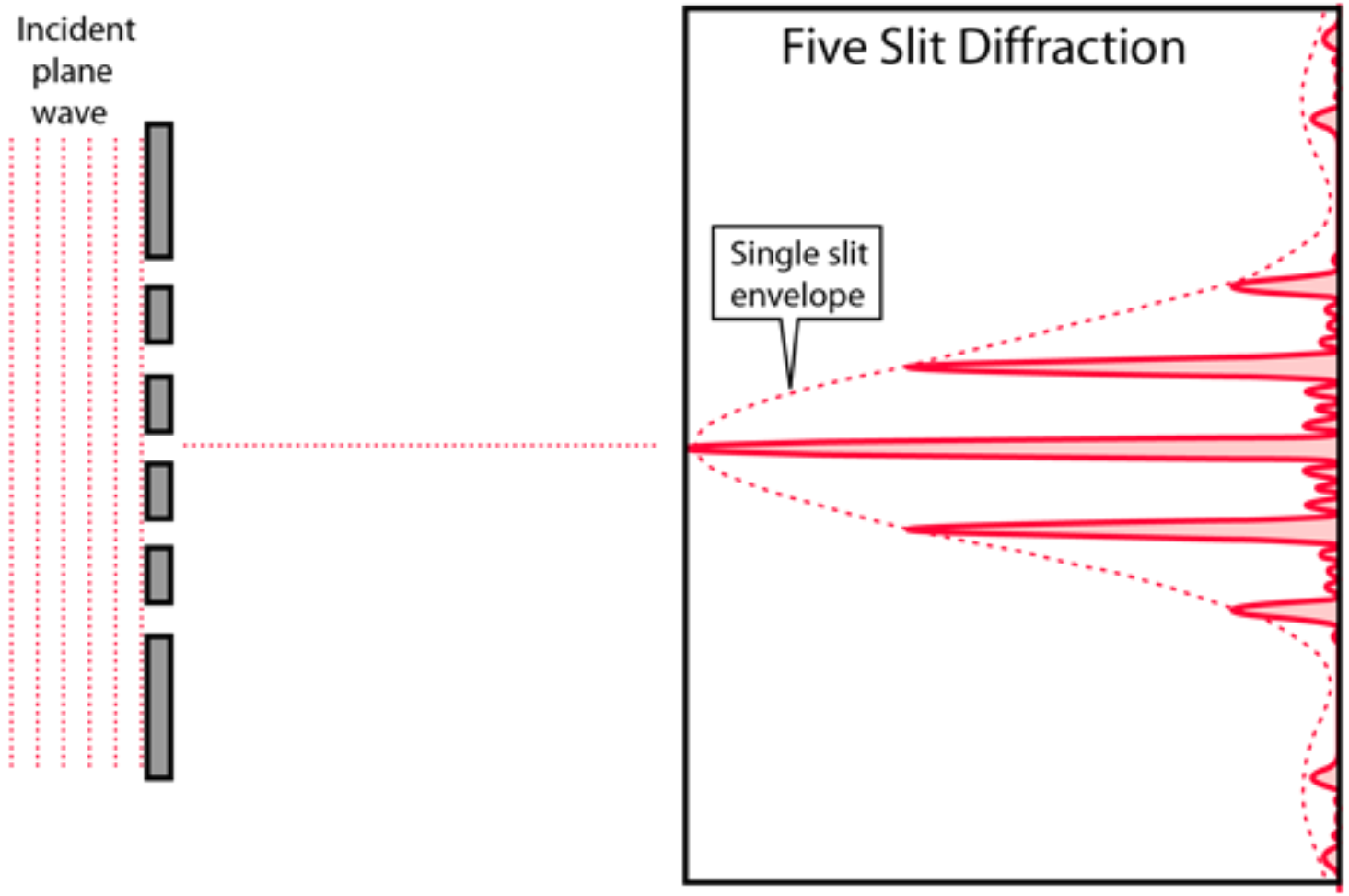

Hyperphysics

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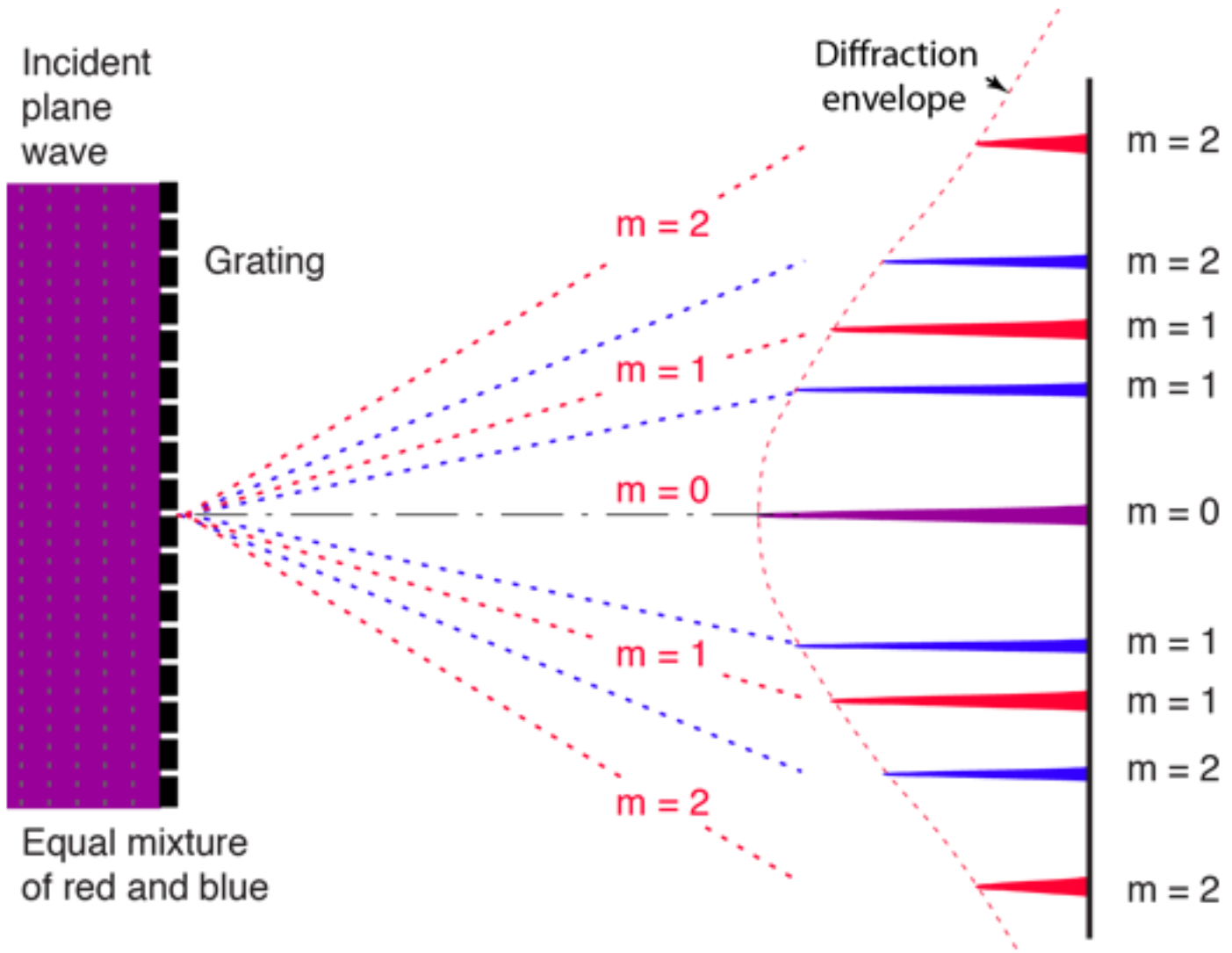

Hyperphysics

# Monochromatic Diffraction



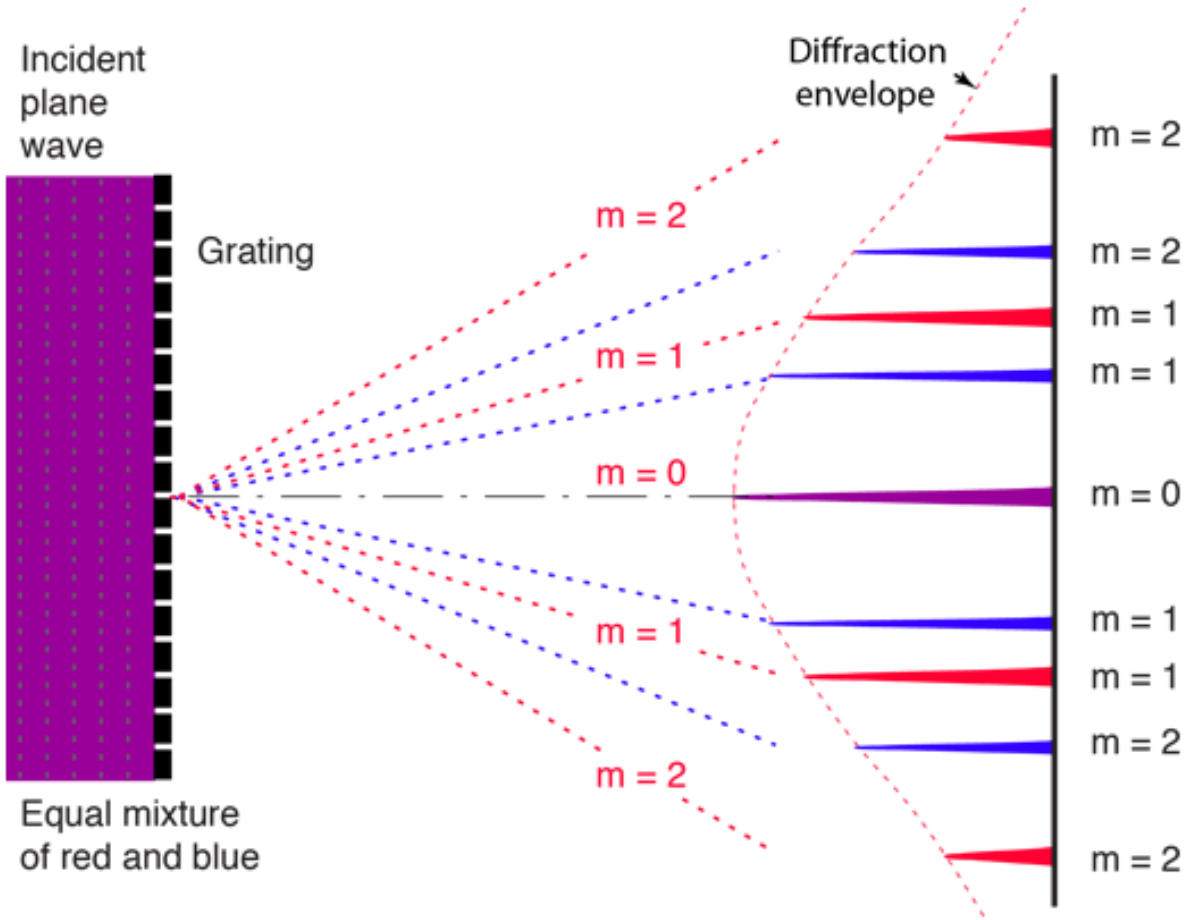

Hyperphysics

# Red + Blue Diffraction

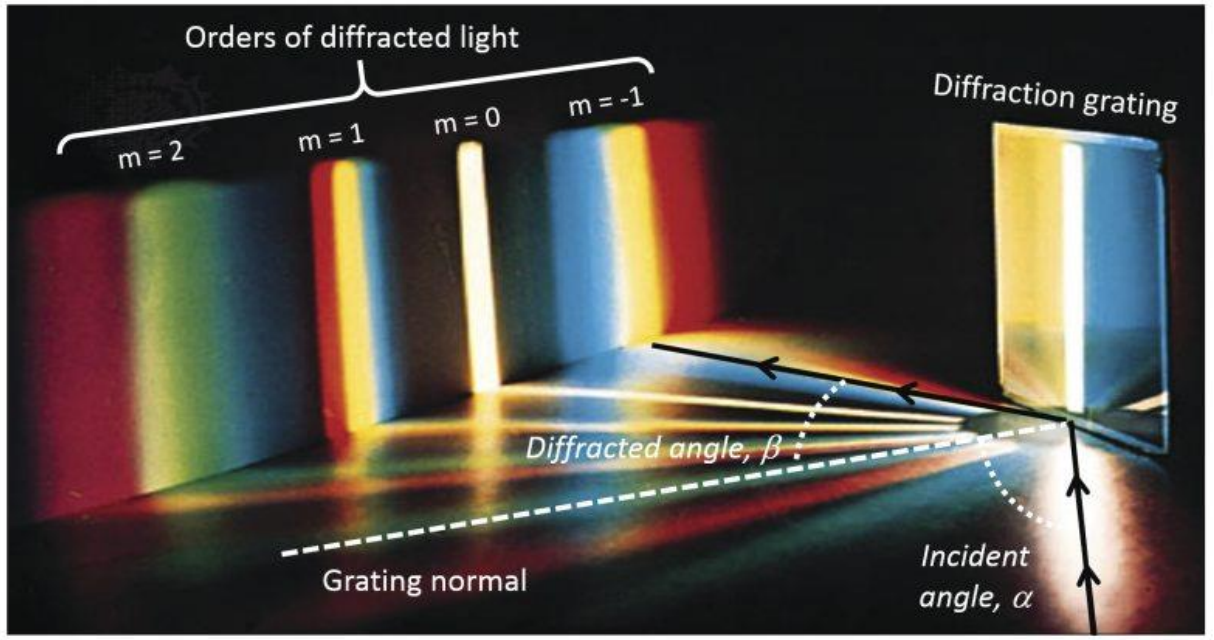



Hyperphysics

# Red + Blue Diffraction

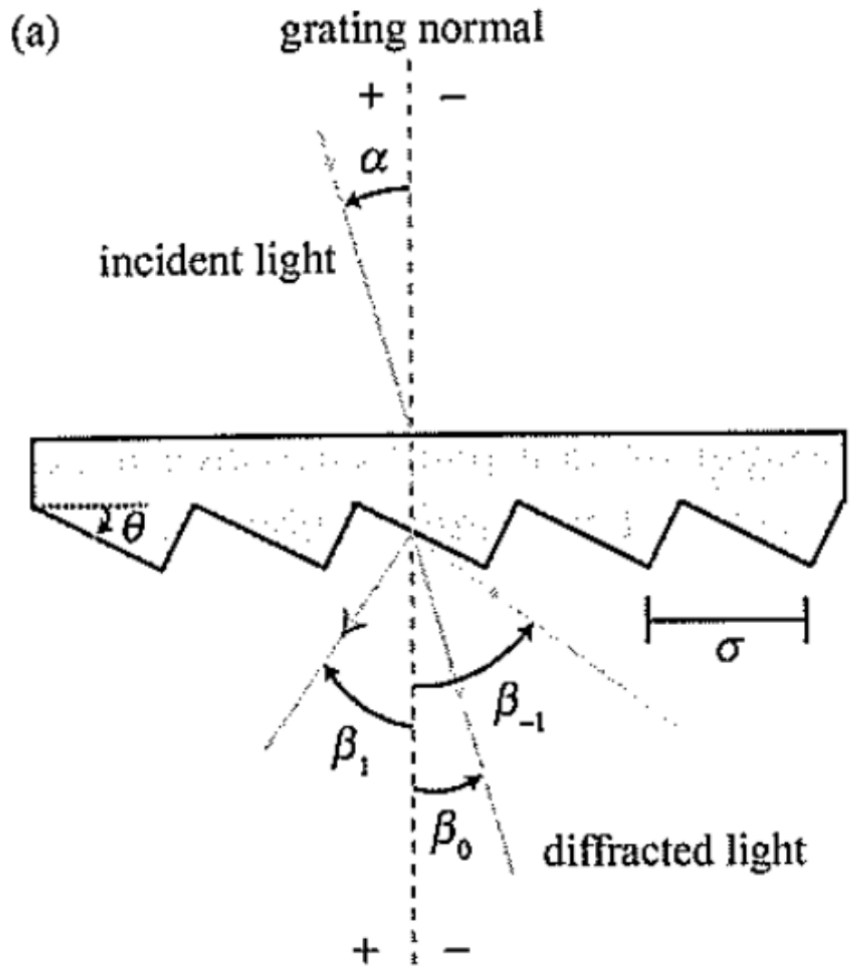


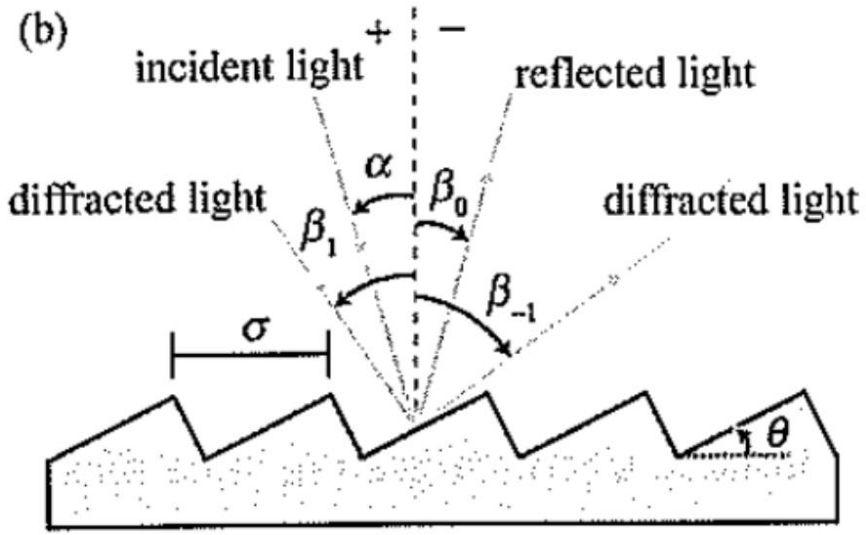

Hyperphysics




Reflective Grating: Newport

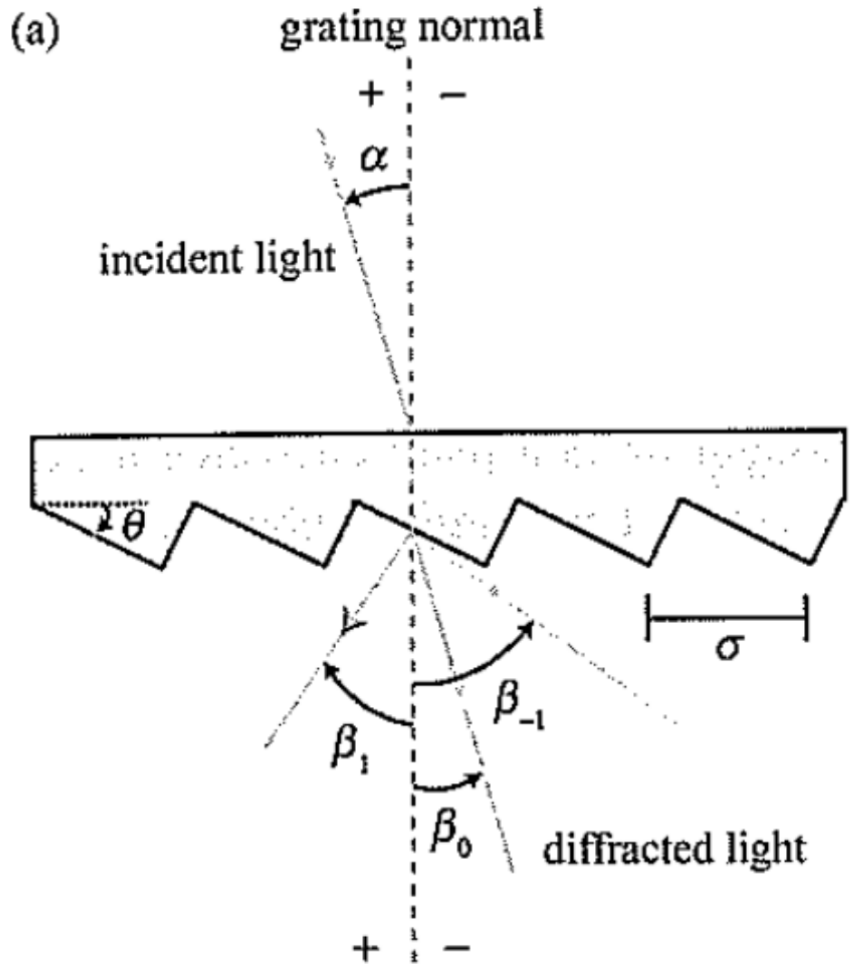
# Grating Parameters




sigma=groove spacing
theta=blaze angle
alpha=incident angle
beta=angle of diffraction

# The Grating Equation



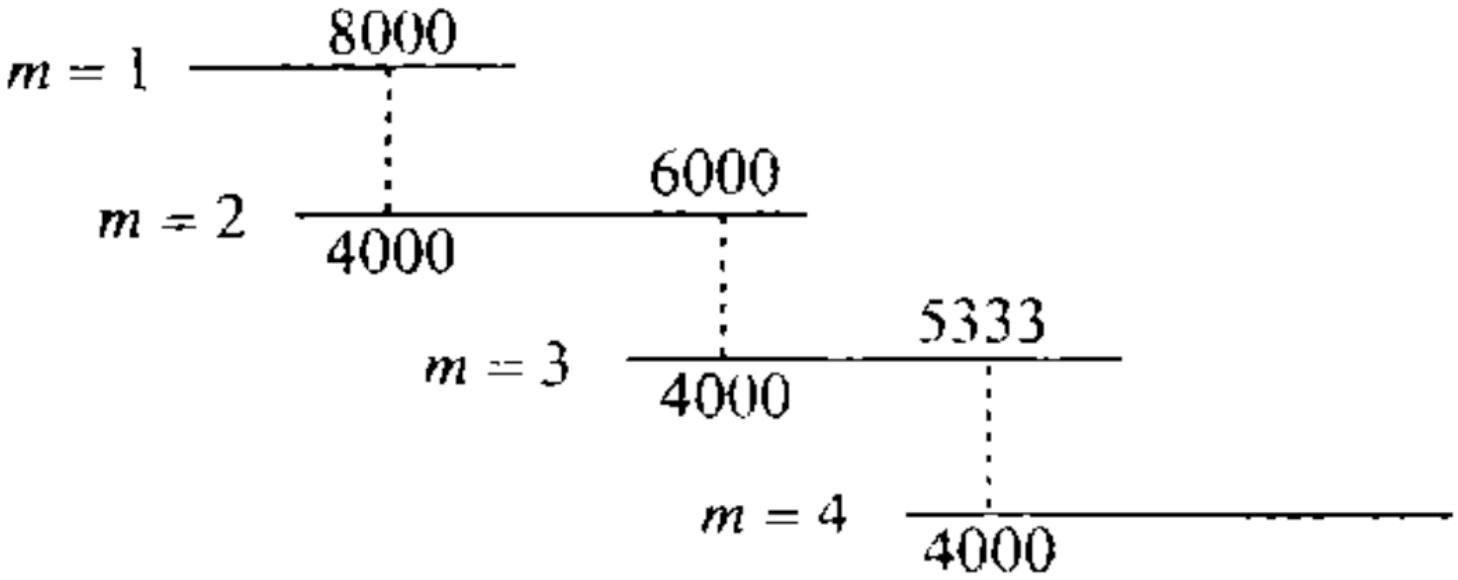

The **Grating Equation** is used to describe how light of a specific wavelength ($\lambda$) will be diffracted at different orders(m). Orders can only be integer values.

Groove or line spacing ($\sigma$) and order determines the grating's dispersive power.

$$m\lambda = \sigma(\sin\alpha + \sin\beta)$$

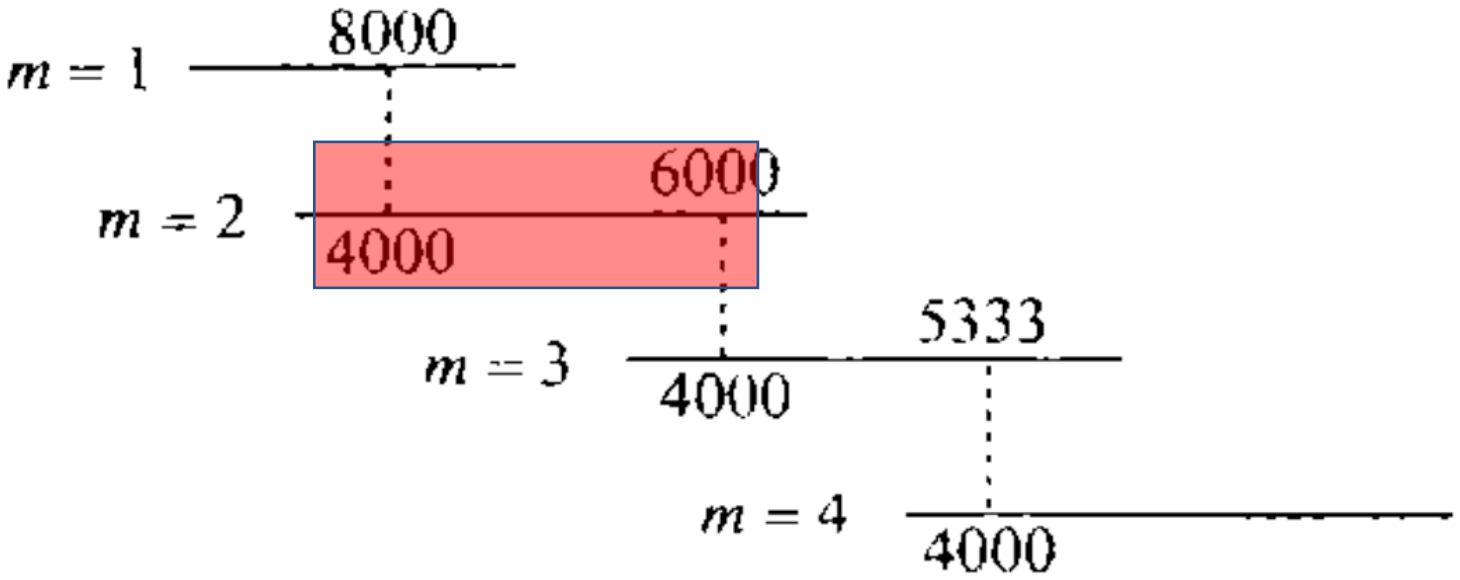
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# Free Spectral Range



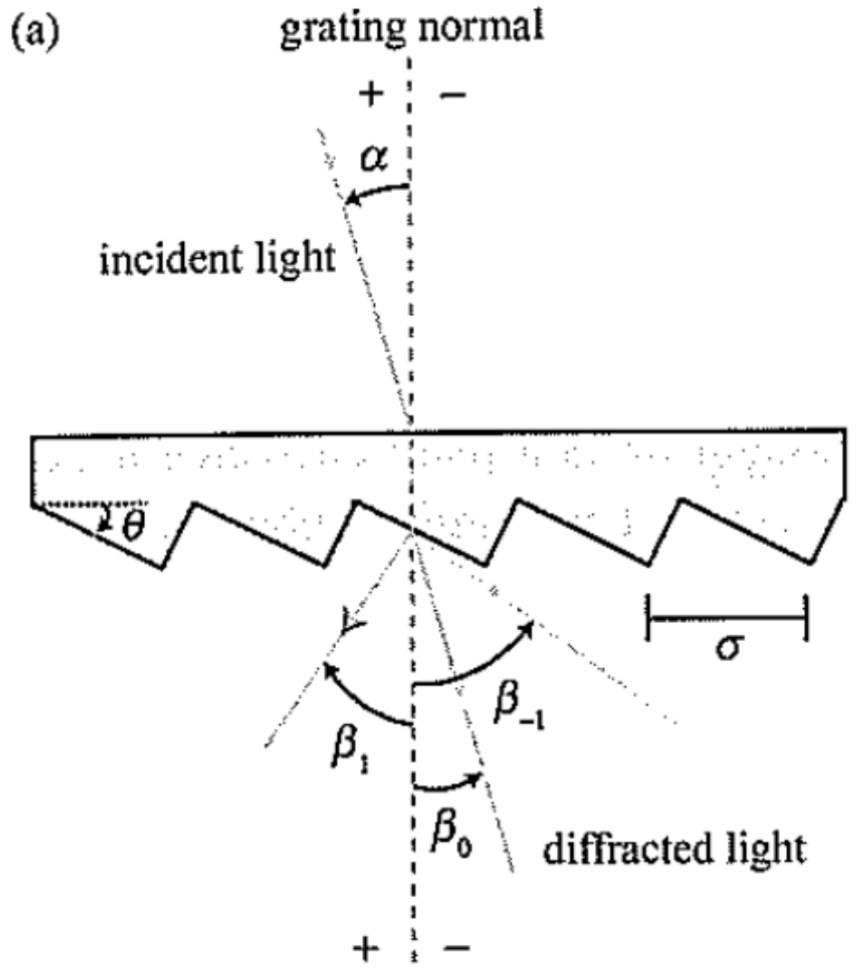

Spectral orders can often overlap, therefore its important to isolate the relevant wavelength range with filters or another mechanism.

# Free Spectral Range




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# Gratings: Echelle vs Echellette

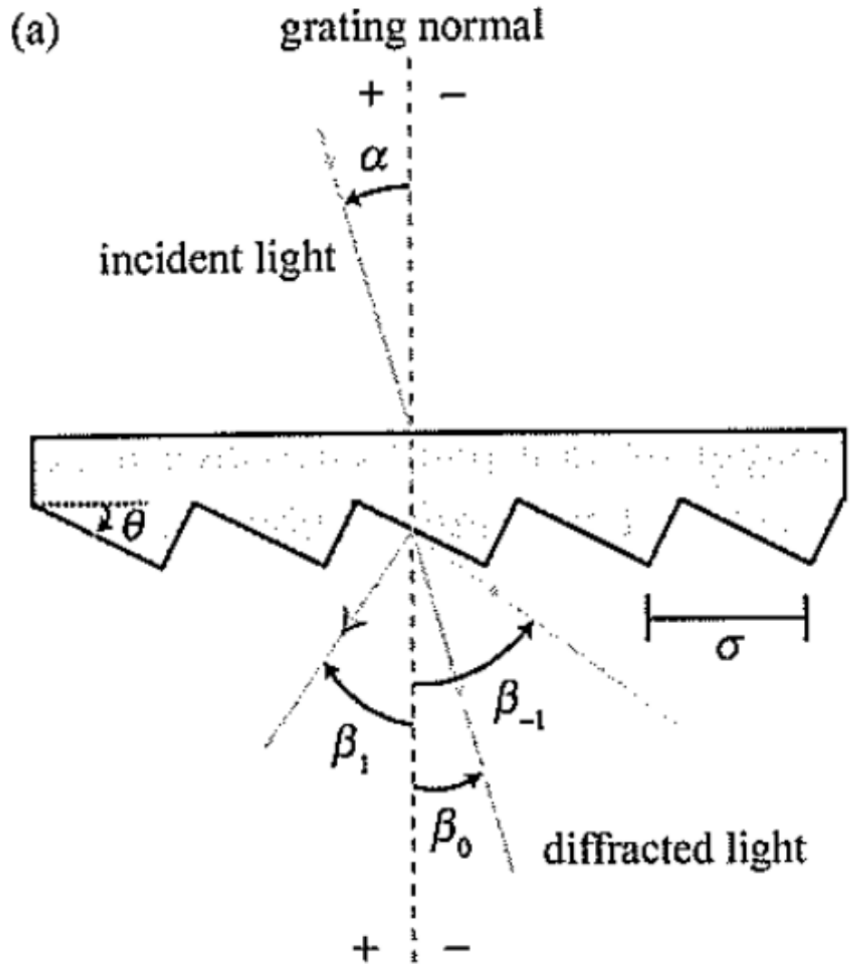



**Echellette gratings** tend to use small groove spacings ($\boldsymbol{\sigma}$, hundreds/mm). Included in instruments at lower incident angles ($\alpha$) for use at low orders.

$$\boldsymbol{m\lambda = \sigma\,(\sin\alpha + \sin\beta)}$$

.

# Gratings: Echelle vs Echellette



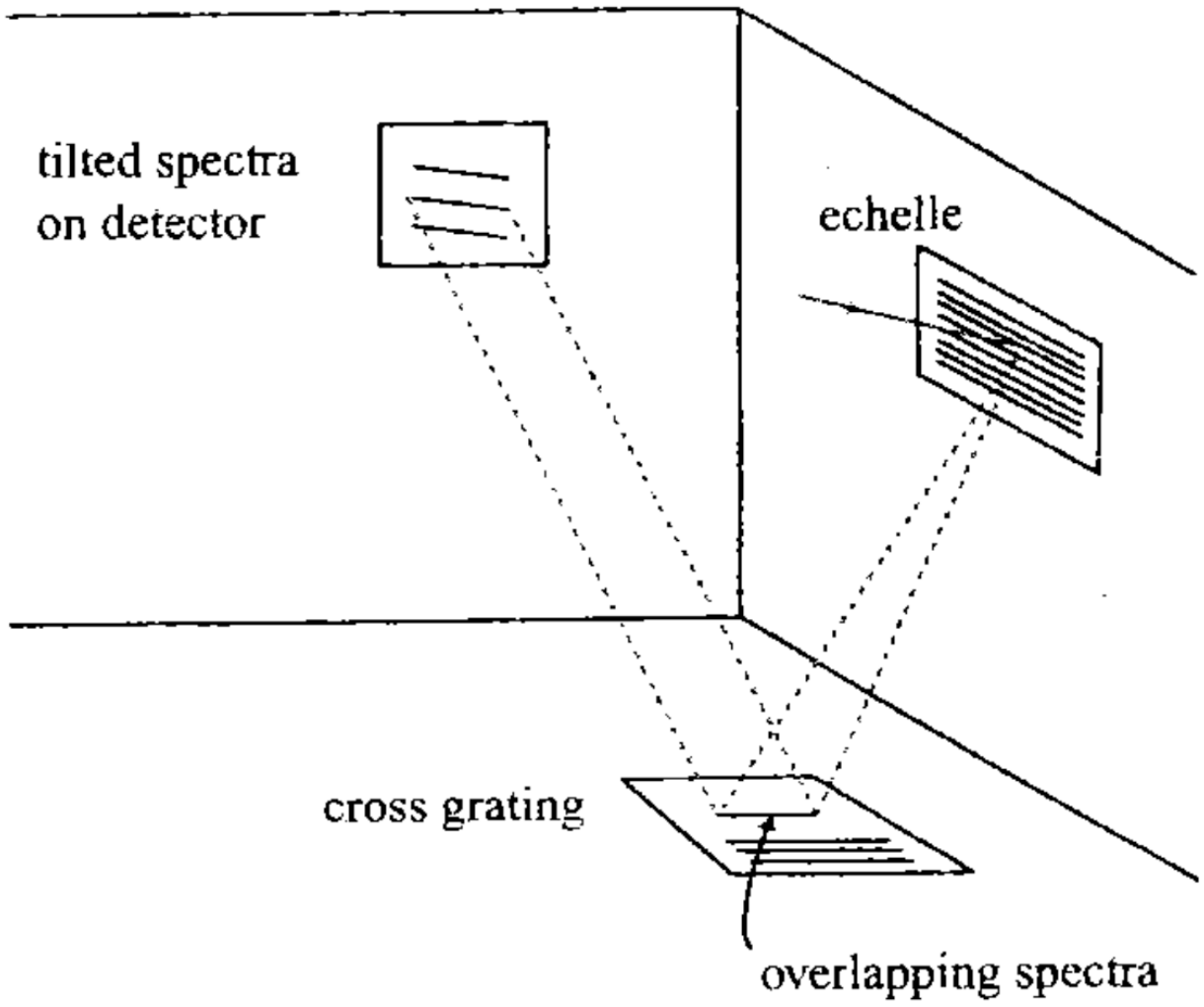

**Echelle gratings** tend to use larger groove spacings ($\boldsymbol{\sigma}$, ten/mm) and steep blaze angles. Included in instruments at high incident angles ($\alpha$) for use at higher orders.

You usually use another grating or disperser behind echelle gratings to separate overlapping orders.

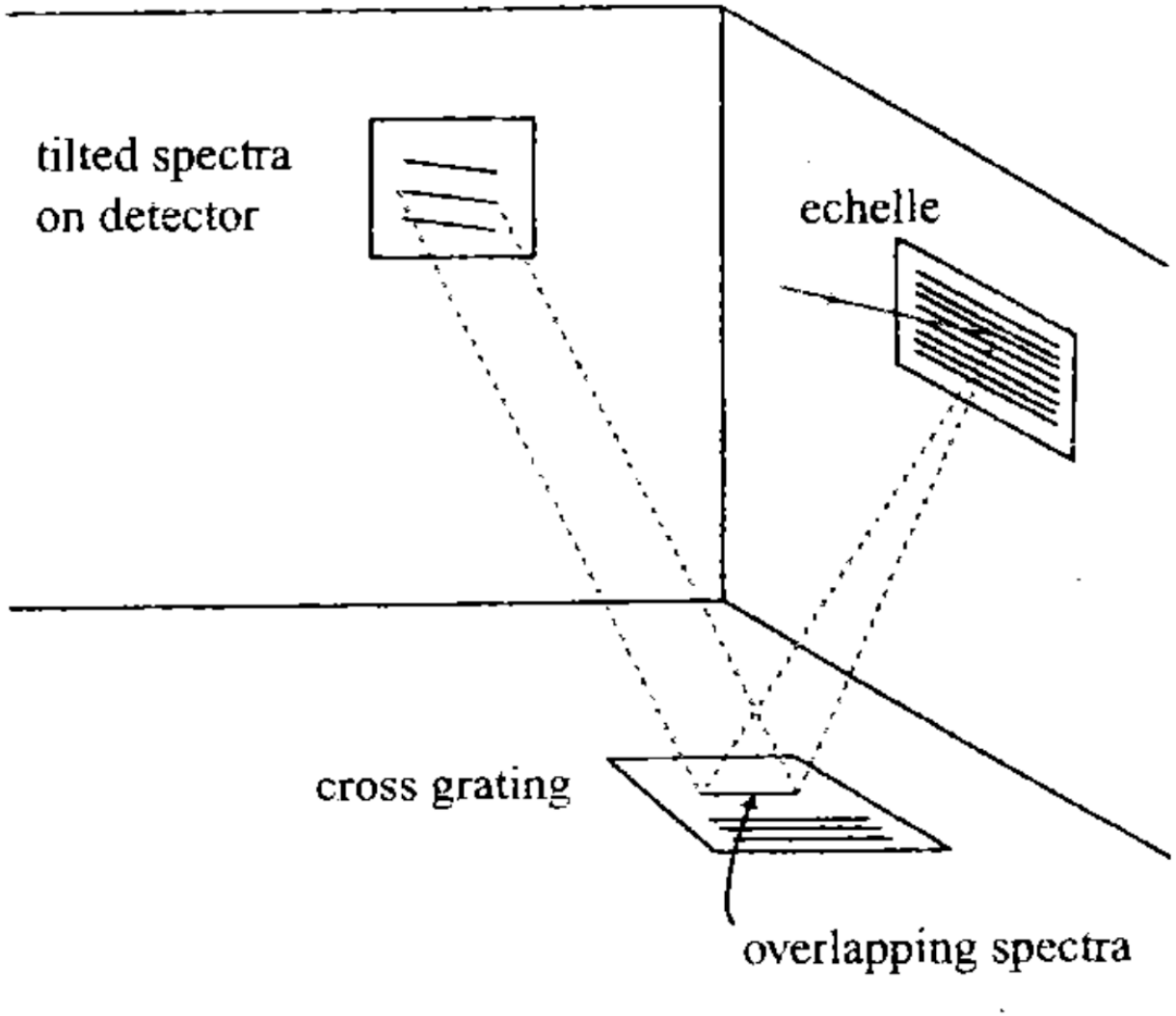
$$\boldsymbol{m\lambda = \sigma\,(\sin\alpha + \sin\beta)}$$

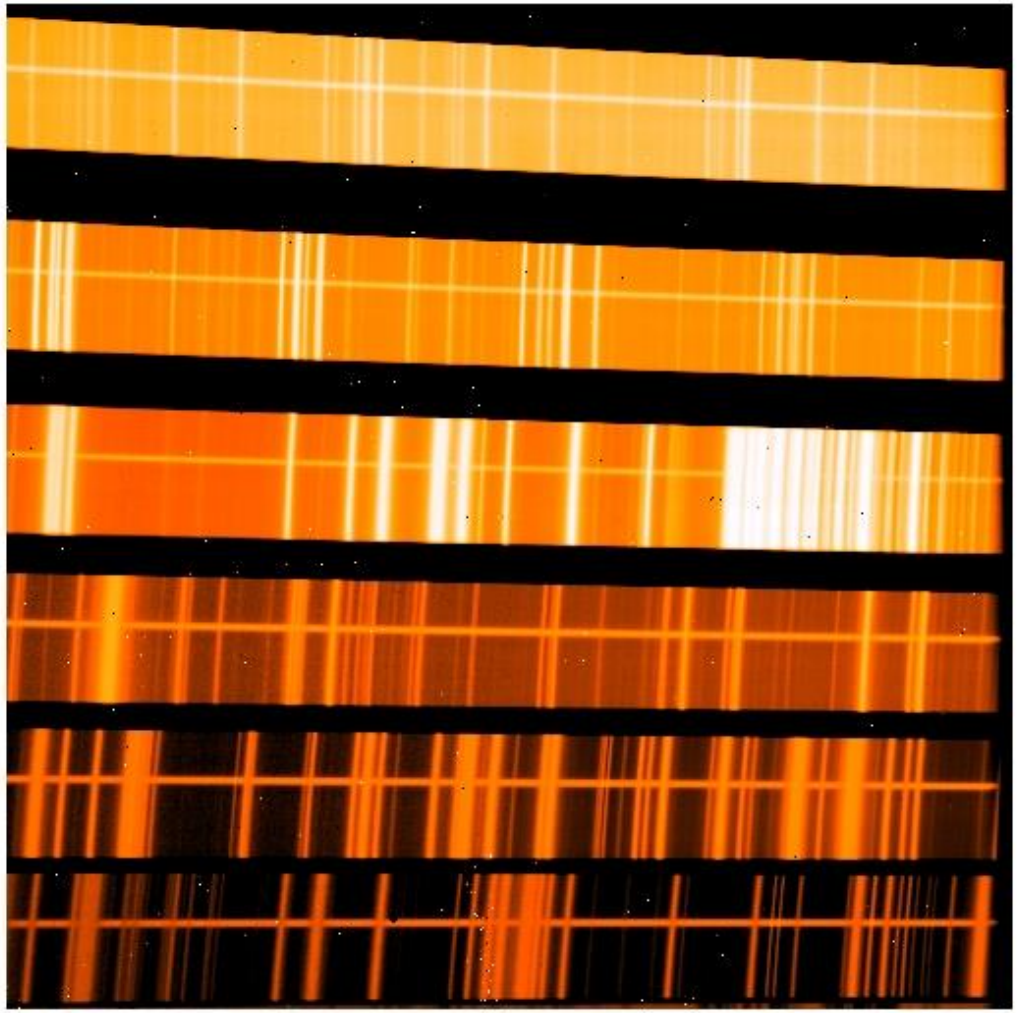
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# Gratings: Echelle Grating with Cross Disperser

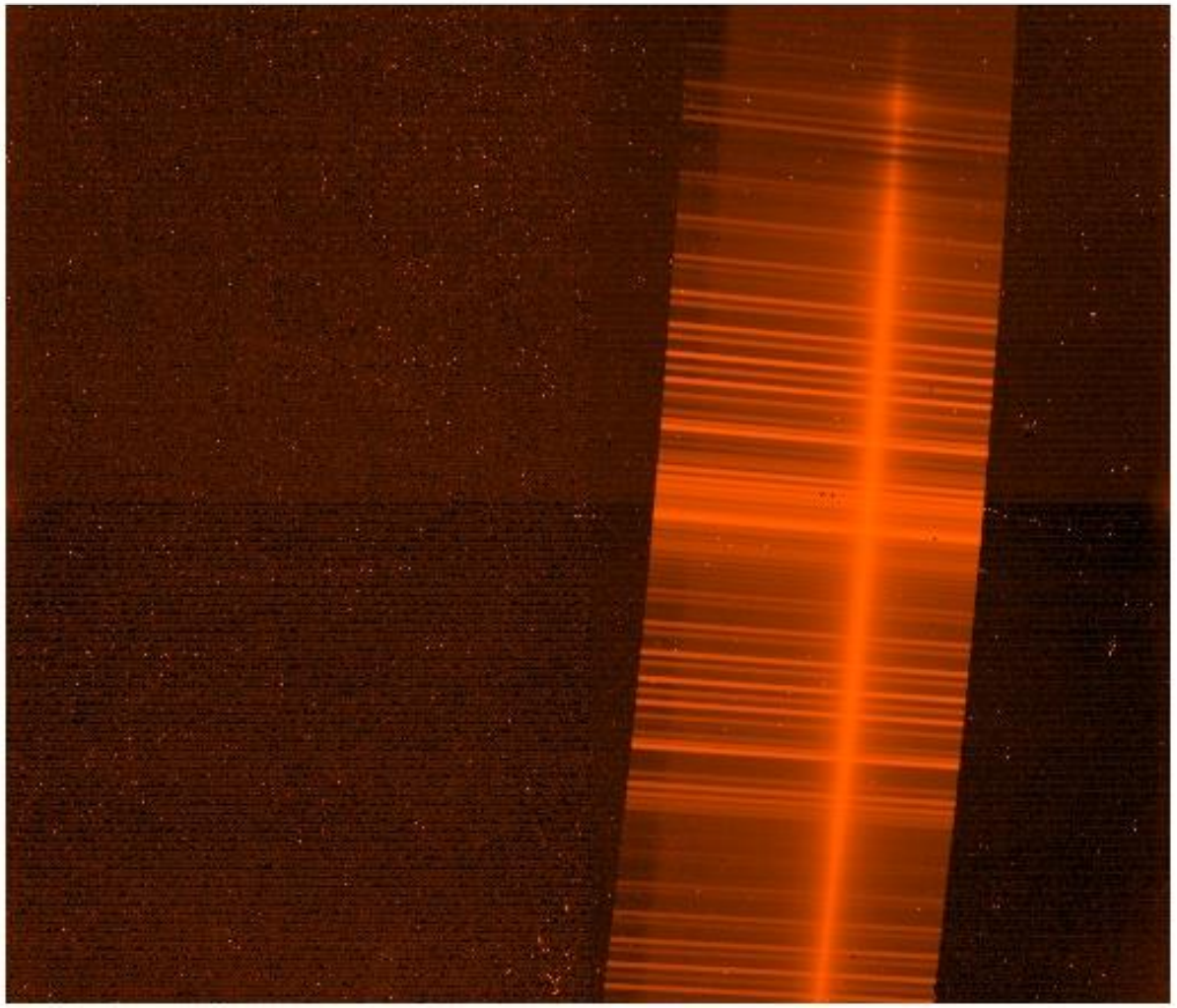


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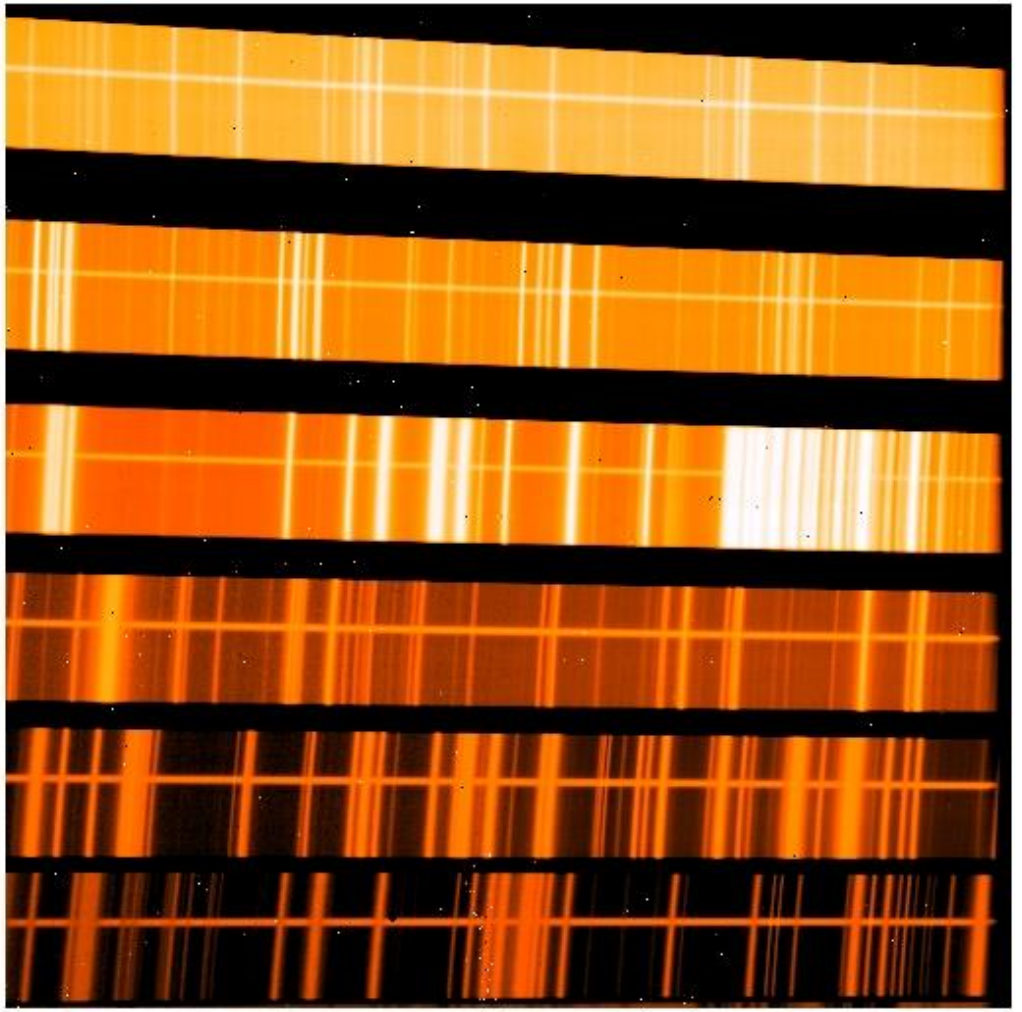


# Keck/NIRSpec

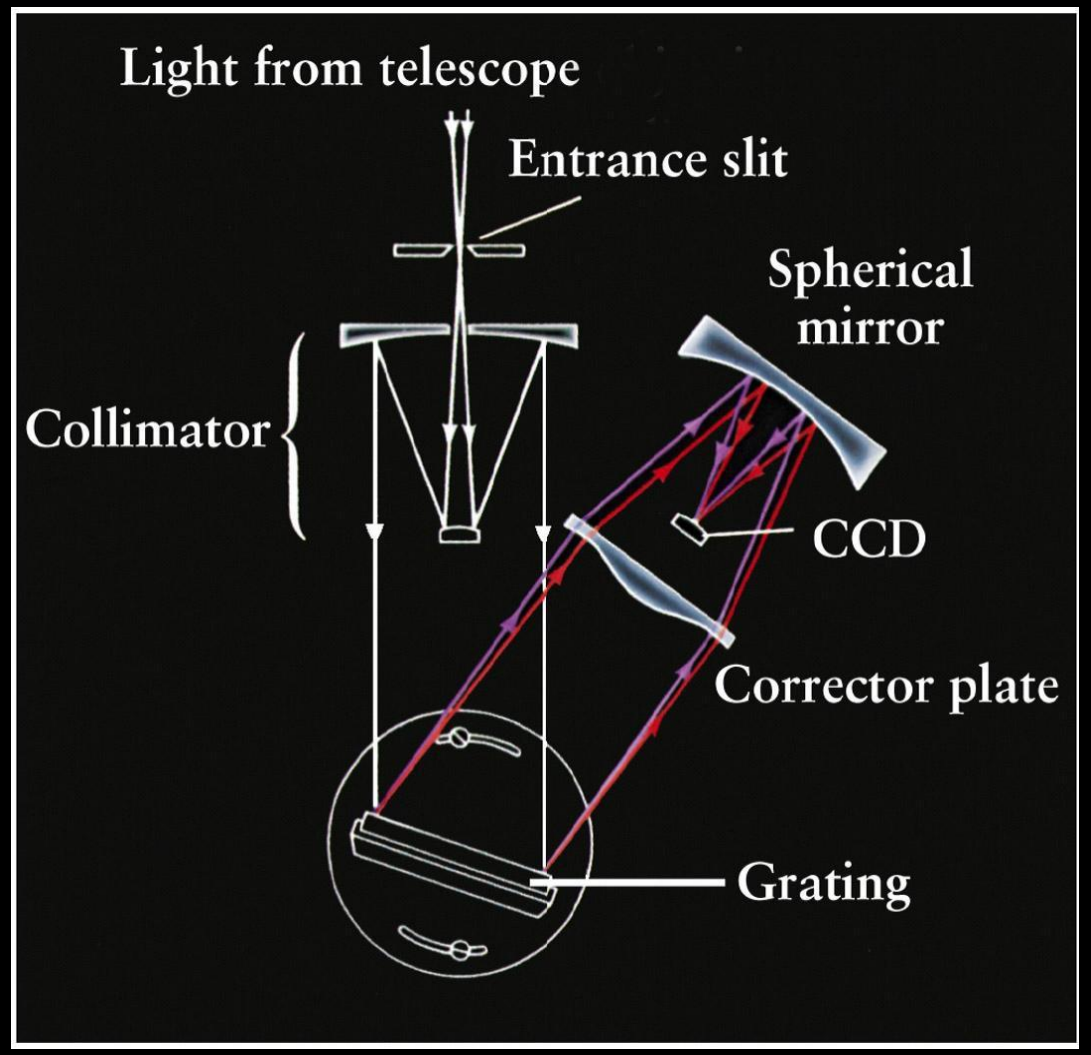


Single Grating (similar to B&C Spectrograph)



Grating + Cross Disperser

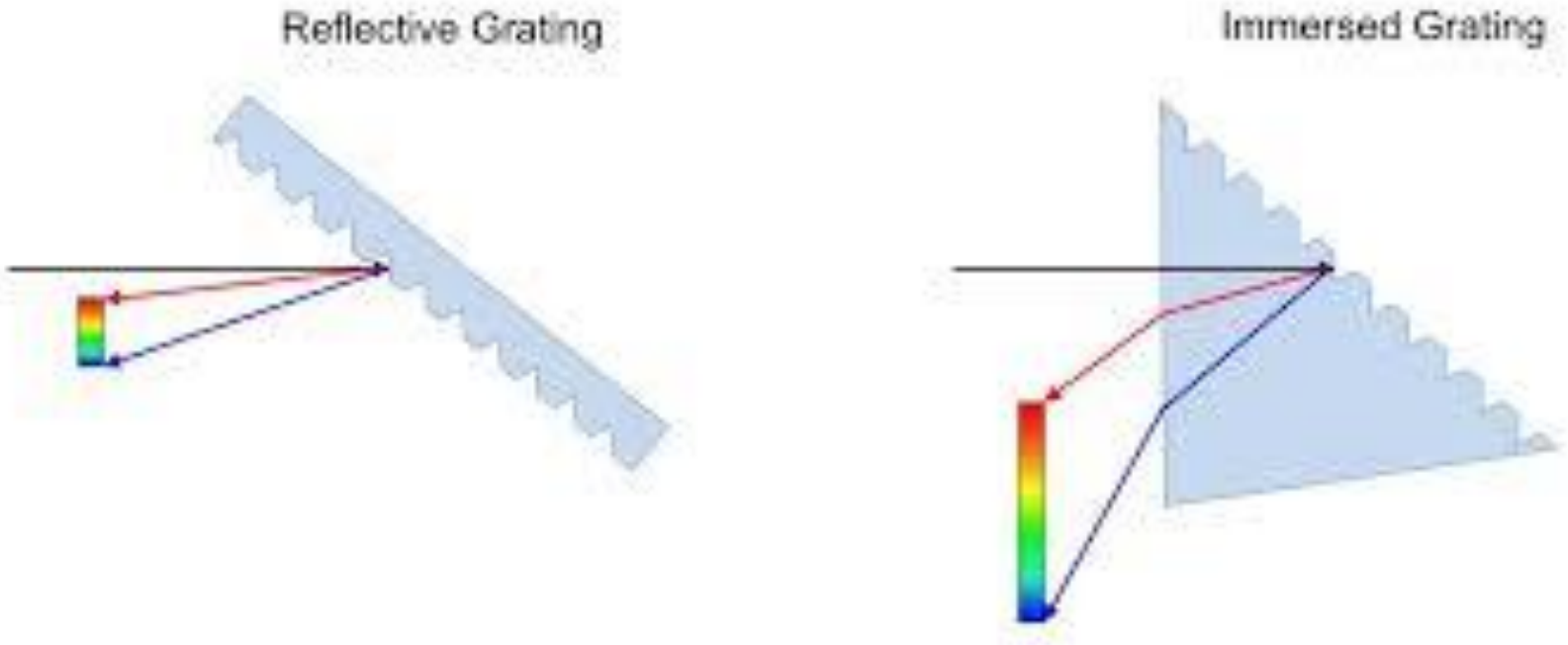
# Components of a Spectrograph with Grating




A reflective grating can be rotated to access different orders and wavelengths with your detector.

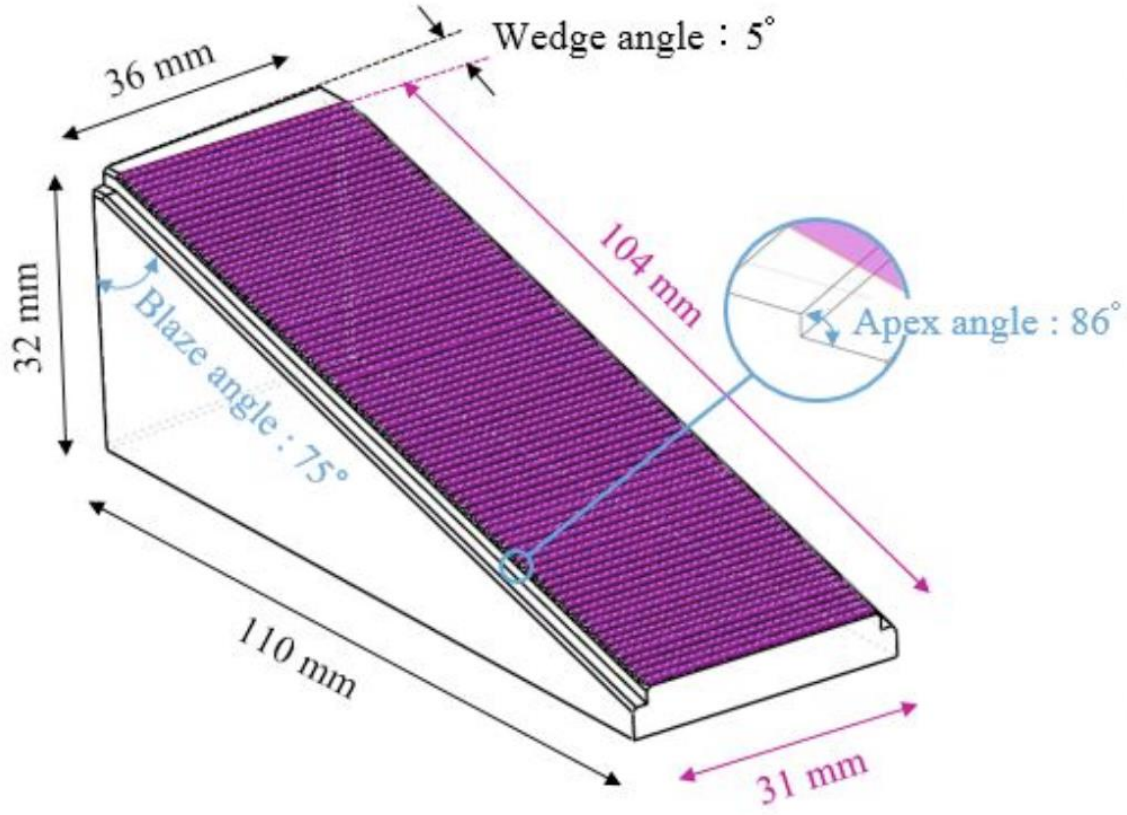
Some instruments keep gratings fixed, but rotate through several to access different wavelengths.

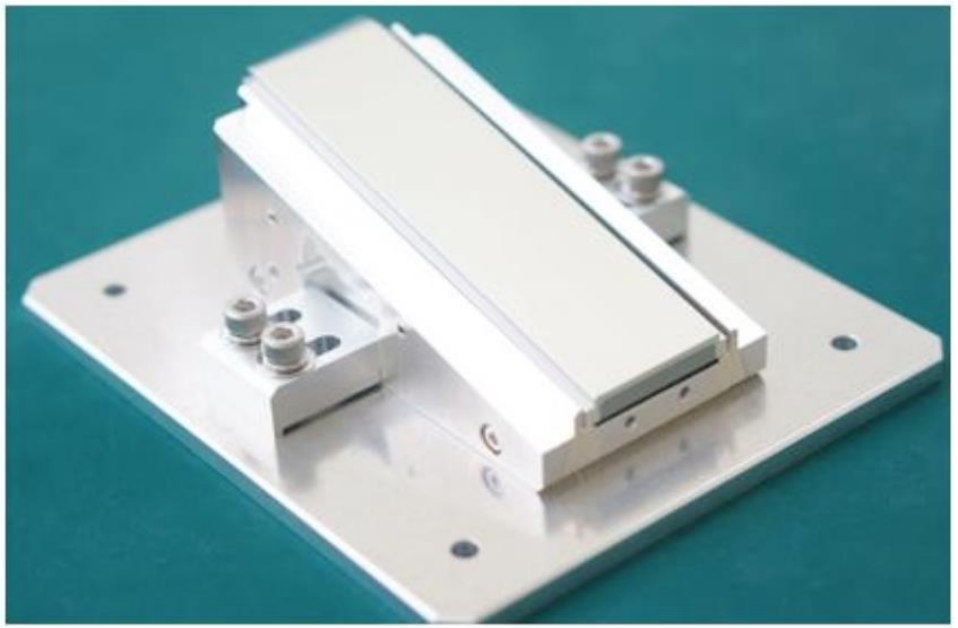
# Other Types of Gratings: Immersion Gratings




**Immersion gratings** are reflective gratings with material of some high index of refraction (n) that can increase the dispersive power of the grating.

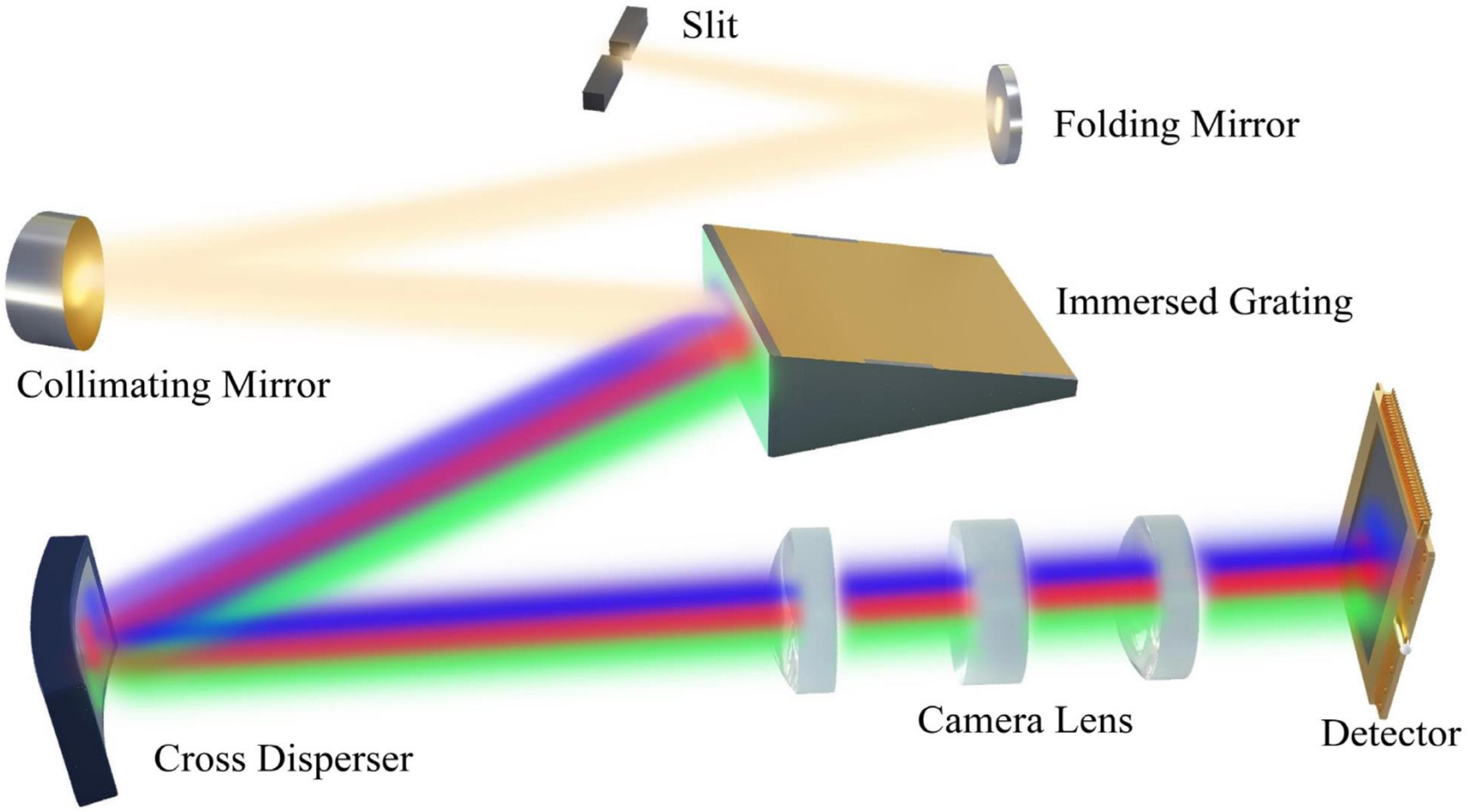
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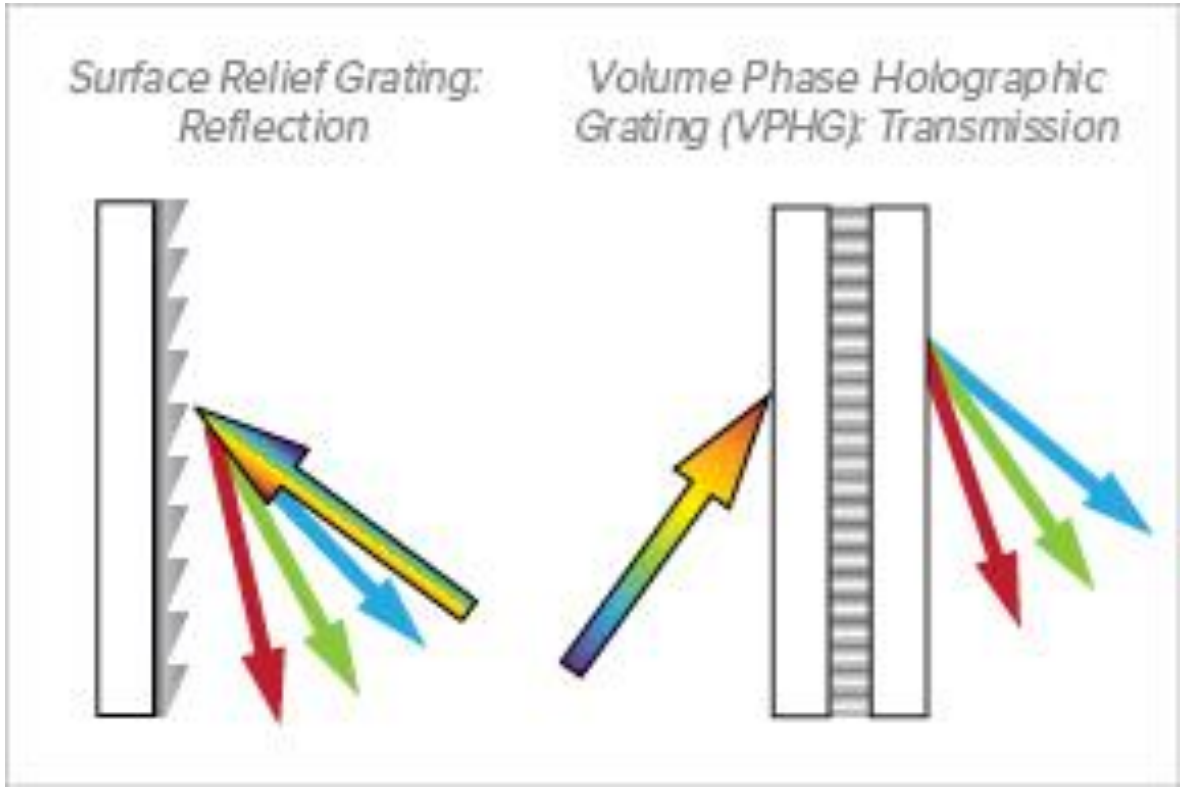
Arasaki+ 2016

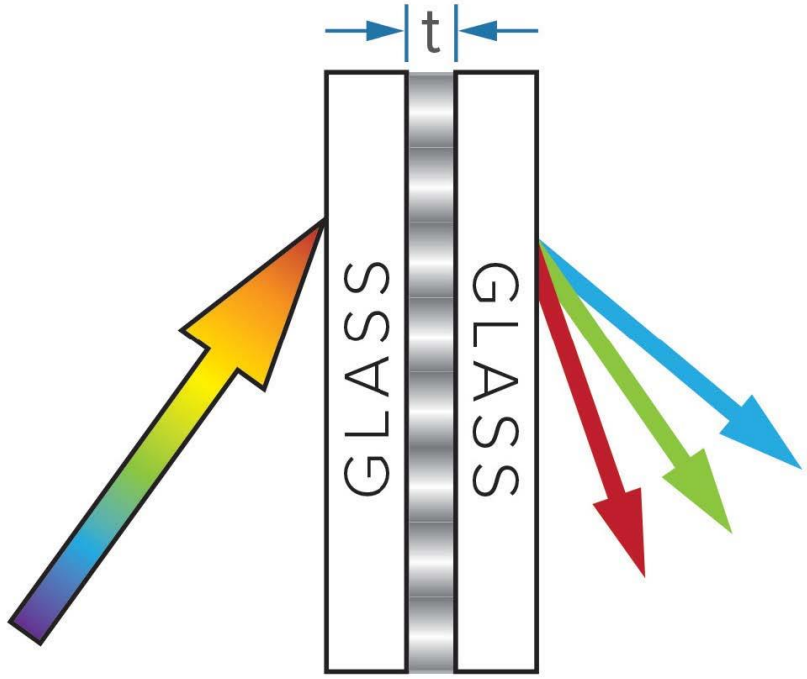
# Other Types of Gratings: Immersion Gratings




Wang+ 2022

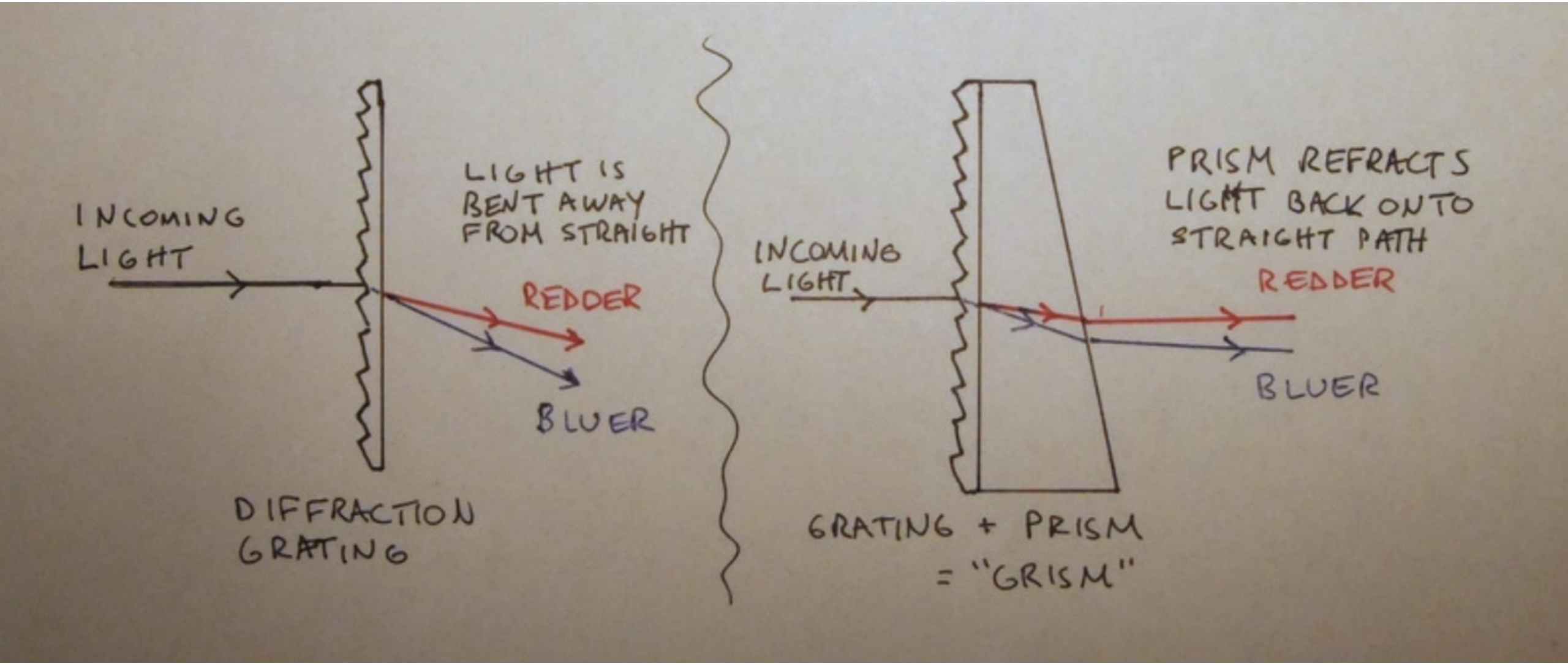
# Other Types of Gratings: Volume Phase Holographic (VPH) Gratings



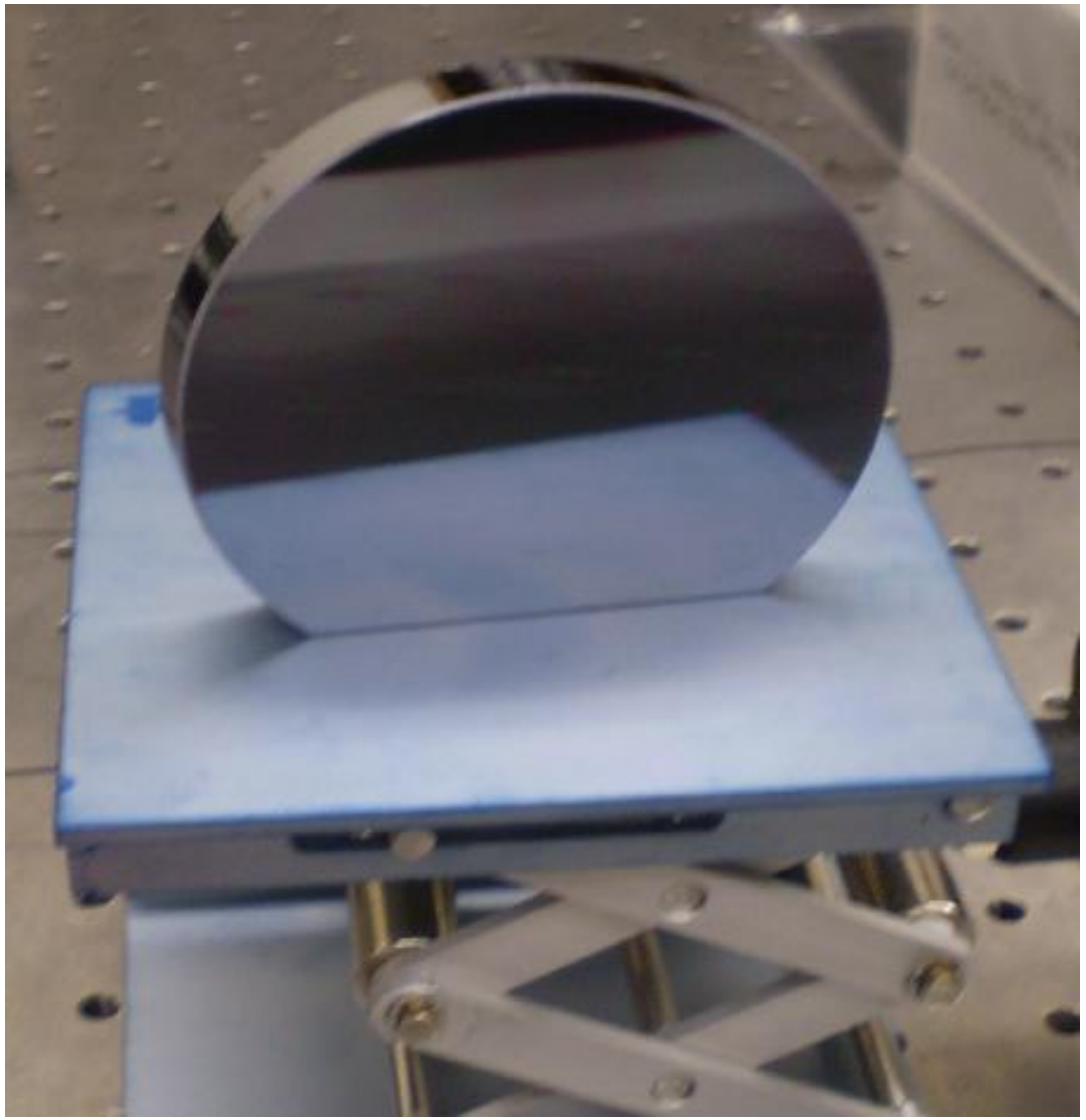

**Volume Phase Holographic Gratings** diffract light by modulating the index of refraction of a dichromated gelatin. Can be reflective or transmissive. Extra reading: Melso+ 2022

# Types of Dispersive Elements: Grisms




From Benjamin Weiner

# Grism Example: JWST/NIRCam

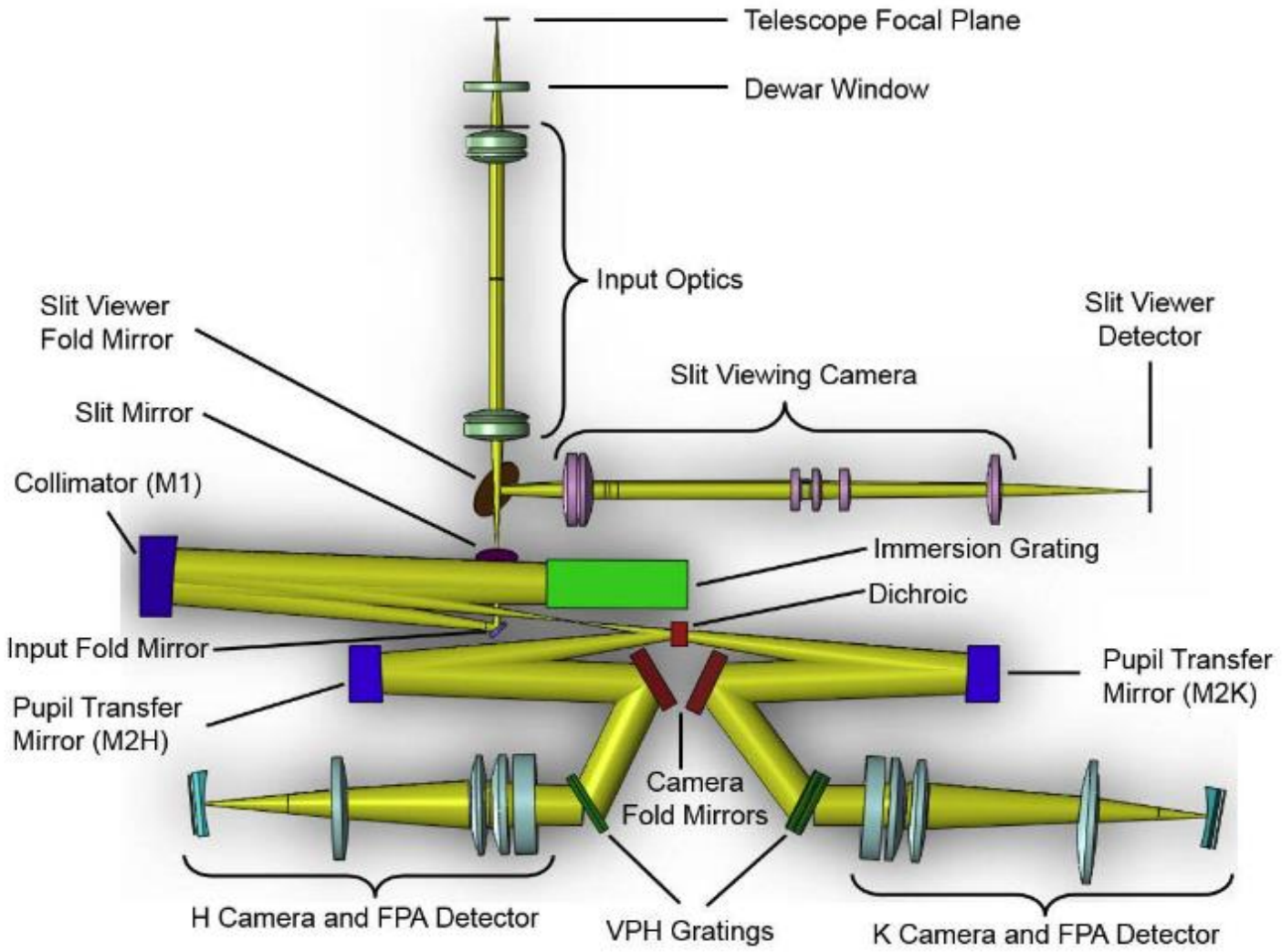


Jaffe+ 2008, Etched grooves into the silicon surface



Jdocs (JWST NIRCam team)

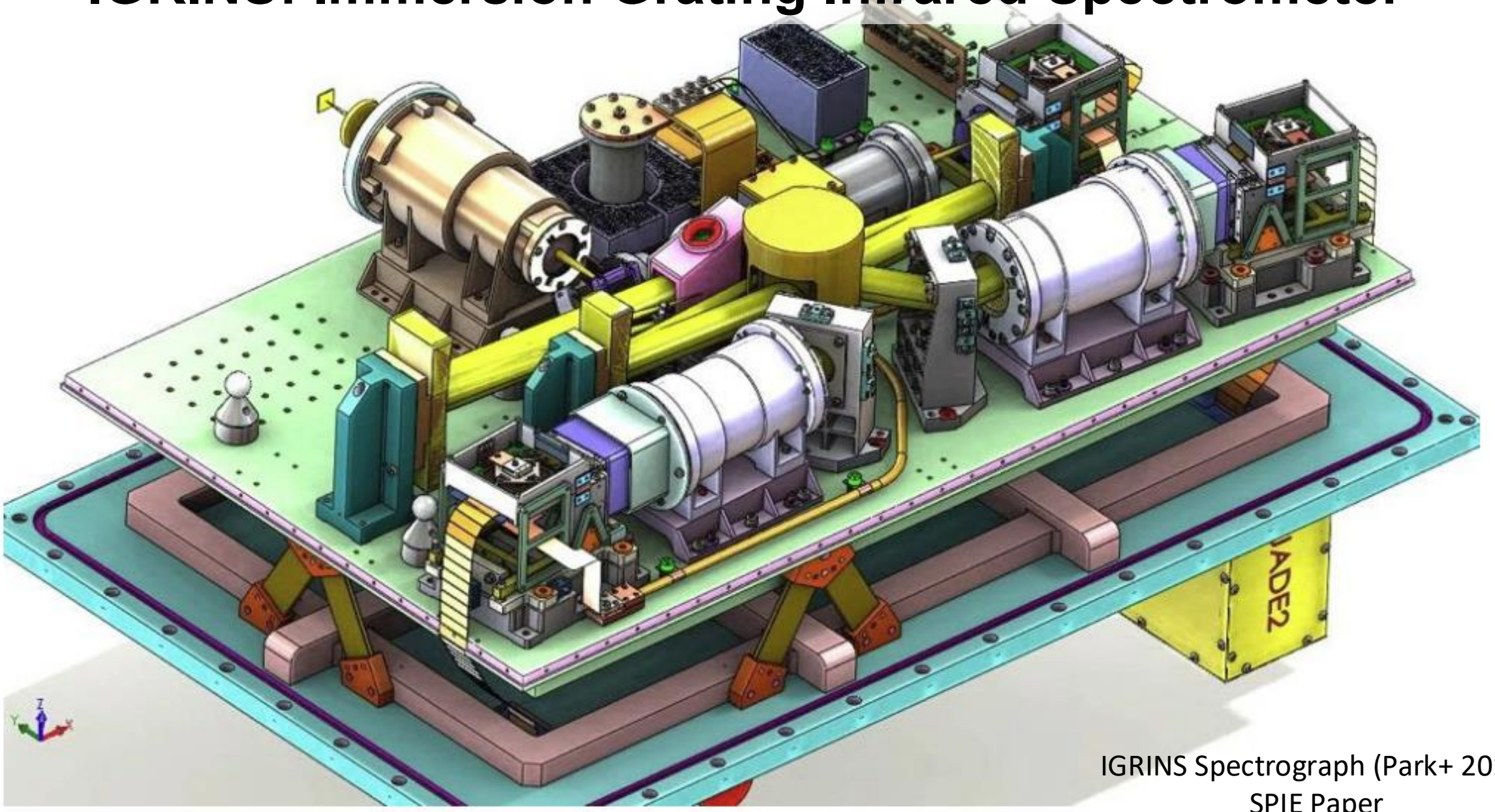
# IGRINS: Immersion Grating Infrared Spectrometer




IGRINS uses both Immersion gratings and VPH gratings

IGRINS Spectrograph (Park+ 2014)
SPIE Paper

# IGRINS: Immersion Grating Infrared Spectrometer



IGRINS Spectrograph (Park+ 2014)
SPIE Paper